

MODULE 1 : EQUITY VALUATION: APPLICATIONS AND PROCESSES

LEARNING OUTCOMES

- 1.a.** Define valuation and intrinsic value and explain sources of perceived mispricing.
- 1.b.** Explain the going concern assumption and contrast a going concern value to a liquidation value.
- 1.c.** Describe definitions of value and justify which definition of value is most relevant to public company valuation.
- 1.d.** Describe applications of equity valuation.
- 1.e.** Describe questions that should be addressed in conducting an industry and competitive analysis.
- 1.f.** Contrast absolute and relative valuation models and describe examples of each type of model.
- 1.g.** Describe sum-of-the-parts valuation and conglomerate discounts.
- 1.h.** Explain broad criteria for choosing an appropriate approach for valuing a given company.

[LOS 1.a] Define valuation and intrinsic value and explain sources of perceived mispricing.

1.

Valuation

Definition: Valuation is the process of determining the value of an asset

Why valuation?

- Stock prices are not perfectly efficient. The goal of valuation is to identify mispriced securities/assets.
- Investment success can depend crucially on the analyst's ability to determine the values of securities.

CTG		BID		VIC
	SHB		PLX	
VRE		GAS		MBB
	SAB		VHM	
MSN		HPG		VPB
	POW		ACB	
SSI		TCB		FPT

Available investments

Asset
valuation →

SSI	VIC
VHM	FPT
TCB	

*Selection of potential
investments*

[LOS 1.a] Define valuation and intrinsic value and explain sources of perceived mispricing.

2.

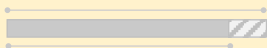
Intrinsic value

Definition: Intrinsic value is the valuation of an asset by someone who has **complete understanding** of the characteristics of the asset or issuing firm.

Intrinsic value role in investing

Analysts seeking to produce positive risk adjusted return by identifying securities which are mispricing by the market. Those are the securities of which **market price differs from intrinsic value**.

Intrinsic value

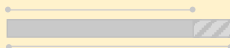


Market price

Intrinsic value > Market price

Potential long position

Intrinsic value



Market price

Intrinsic value < Market price

Potential short position

[LOS 1.a] Define valuation and intrinsic value and explain sources of perceived mispricing.

3.

Perceived mispricing

Mispricing perceived by the analyst into two sources:

- The difference between market price and the intrinsic value (actual mispricing)
- The difference between the analyst's estimate of intrinsic value and actual intrinsic value (valuation error).

Market price



IV_{Actual}



$IV_{Analyst}$



=



+



**Mispricing perceived
by analyst**

Actual mispricing

Valuation error

$IV_{Analyst} - \text{price}$ =

$(IV_{Actual} - \text{price})$ + $(IV_{Analyst} - IV_{Actual})$

[LOS 1.b] Explain the going concern assumption and contrast a going concern value to a liquidation value.

1.

Going concern assumption

Definition: **Going concern assumption** is the assumption that a company will continue to operate as a business in a foreseeable future. *A lot of assets only have value if the going concern assumption is prevailed.*



Going concern assumption



Models of going-concern value are our focus.

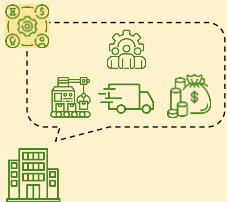
Some implication of going concern assumption:

- Company continues to produce and sell its goods and services;
- Company uses its assets in a value-maximizing way for a relevant economic time frame;
- Company accesses its optimal sources of financing.

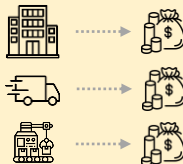
[LOS 1.b] Explain the going concern assumption and contrast a going concern value to a liquidation value.

2.

Going concern value vs. Liquidation value



If a firm is under a going-concern assumption, its value will be **going concern value**.



If a firm is under financial distress, a going concern assumption may not be appropriate. The value if it were dissolved and its assets sold individually, known as its **liquidation value**.

? Why going concern value differs from liquidation value?

- For many companies, the value added by assets working together and by human capital applied to managing those assets makes estimated going-concern value greater than liquidation value.
- A persistently unprofitable business may be worth more “dead” than “alive”. In such case, liquidate the company is the optimal choice.

[LOS 1.c] Describe definitions of value and justify which definition of value is most relevant to public company valuation.

1.

Fair market value

Definition: Fair market value is the price at which a hypothetical willing, informed, and able seller would trade an asset to a willing, informed, and able buyer.

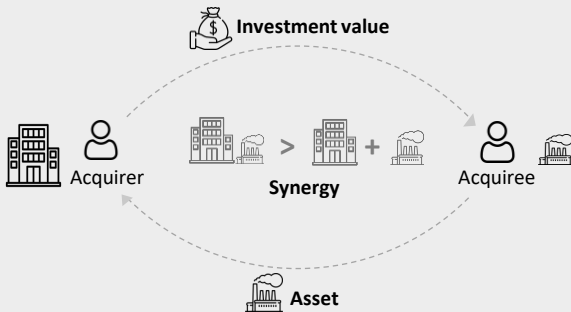


[LOS 1.c] Describe definitions of value and justify which definition of value is most relevant to public company valuation.

2.

Investment value

Definition: Investment value is the value of a stock to a particular buyer. Investment value may depend on the buyer's specific **needs and expectations**, as well as perceived **synergies** with existing buyer assets.



[LOS 1.d] Describe applications of equity valuation.

1.

The process of valuation



Understand

the business
model



Forecast

company
performance



Select

valuation
model



Convert

forecast
into a
valuation



Apply

the
valuation
conclusion



After the valuation process, analysts can determine the current worth of a business or a share of a company.

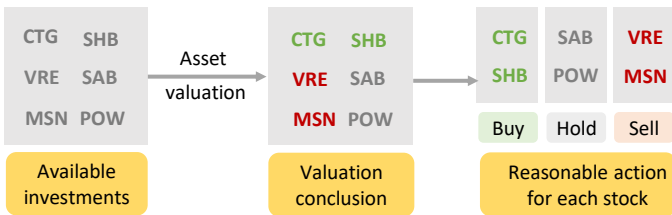
[LOS 1.d] Describe applications of equity valuation.

2.

Valuation – The objectives

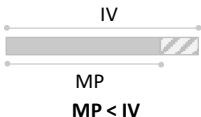
1. Stock selection

The most direct use of equity valuation is to guide the purchase, holding, or sale of stocks.



2. Reading the market

Assume price reflects consensus expectations of investors about future performance, analysts can estimate these expectations by comparing market price with a stock's intrinsic value.



? What is the market expecting about company's future?



Earning growth may slowdown in the next quarter.

[LOS 1.d] Describe applications of equity valuation.

2.

Valuation – The objectives

3. Projecting the value of corporate actions

Market professionals use valuation techniques to determine the result of corporate events (*ex: mergers, acquisitions, divestitures,...*).

Example: Valuation is used to estimate the value of a company after a corporate merger.

 +  =

\$5 billion \$2 billion

 ? billion

Valuation to estimate the effect of a M&A

→ Support decision making
→ Guidance for market participants

4. Fairness opinions

Analysts use equity valuation to support professional opinions about the fairness of a price to be received by minority shareholders in a merger or acquisition.



Minority
shareholders
in a M&A



Valuation
process to
support opinion



Conclusion
about fairness
of the price

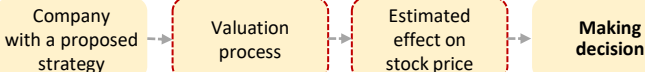
[LOS 1.d] Describe applications of equity valuation.

2.

Valuation – The objectives

5. Planning and consulting

Many firms engage analysts to evaluate the effects of proposed corporate strategies on the firm's stock price.



Valuation made by analyst

6. Communication with analysts and investors

Valuation provide a common basis for management, investors, and analysts to discuss the company's performance, current state, and future plans.



Different views



Valuation process
aligns views and
ideas



Common basis to
discuss

[LOS 1.d] Describe applications of equity valuation.

2.

Valuation – The objectives

7. Valuation of private business

Analysts use valuation techniques to determine the value of firms or holdings in firms that are **not publicly traded** for multiple financial purposes (transactional, taxation, IPO,...)

8. Portfolio management

The investment process is usually considered to have three parts: planning, execution, and evaluation of results. Equity valuation is a primary concern in the first two of these steps.

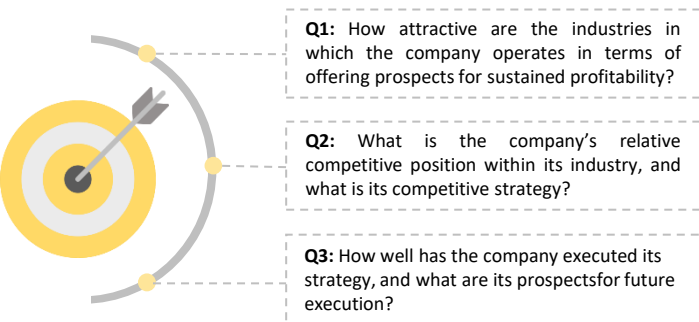
-  **Step 1:** Planning
-  **Step 2:** Executing the investment plan
-  **Step 3:** Evaluation of results



Equity valuation is a primary concern in the first two steps.

[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

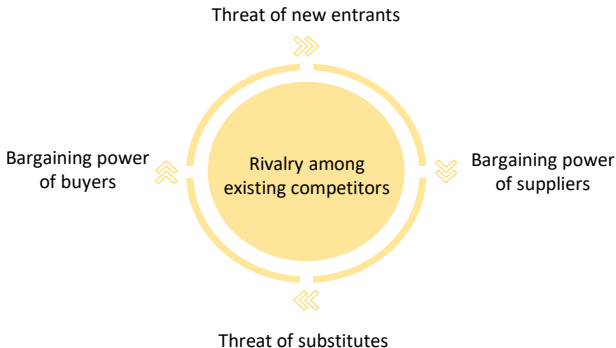
- The content of this LOS explains detailedly the first step of the valuation process – Understand the business model.
- When conducting an industry and competitive analysis, it's important to answer 03 questions:



[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

? Question 1: How attractive are the industries in which the company operates in terms of offering prospects for sustained profitability?

 The attractiveness (long-term profitability) of any industry is determined by the interaction of these five competitive forces (Porter's five forces).



[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

? Question 2: What is the company's relative competitive position within its industry, and what is its competitive strategy?

- A company's value is higher to the extent that it can create and sustain an advantage relative to its competition.
- There are several generic corporate strategies for achieving above-average performance:



Cost leadership

Being the lowest cost producer while offering products comparable to those of other companies so that products can be **priced at or near the industry average**.



Differentiation

Offering unique products or services along some dimensions that are widely valued by buyers so that the company can **command premium prices**



Focus

Seeking a competitive advantage within a target segment or segments of the industry based on either cost leadership (cost focus) or differentiation (differentiation focus)

Once the analyst has identified a company's strategy, he can evaluate the performance of the business over time in terms of how well it executes its strategy and how successful it is.

[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

? Question 3: How well has the company executed its strategy, and what are its prospects for future execution?

To evaluate a company's performance against its strategic objectives and develop expectation about its future performance, analyst can analyze the company's past financial reports.



Company's strategy



Result reflected in past
financial report

- Many inputs of valuation models come from accounting information in the firm's financial reports.
 - In order for the analyst to successfully estimate the value of the firm, the financial factors must be disclosed in sufficient detail and accuracy.
 - The analyst must also consider the reporting quality of the firm.

[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

? Is the financial statement information reported in good quality?

Quality of earnings issues can be broken down into several categories and may be addressed only in the footnotes and disclosures to the financial statements.

1. Accelerating or premature recognition of income

2. Reclassifying gains and nonoperating income

3. Expense recognition and losses

4. Amortization, depreciation, and discount rates

5. Off-balance-sheet issues

[LOS 1.e] Describe questions that should be addressed in conducting an industry and competitive analysis.

? Is the financial statement information reported in good quality?

Warning signs of poor earnings quality (and risk of negative earnings surprises)

1. Past history of SEC violations, or late filings.
2. Related-party transactions.
3. Excessive loans to officers, employees, or directors.
4. Poor accounting disclosures related to (for example) segmental information or accounting assumptions and policies, and inadequate discussion of negative factors.
5. High management or director turnover.
6. Consulting services provided by an audit firm.
7. Disputes with and/or changes in auditors.
8. Executive compensation tied to profitability or stock price.
9. Declining margins or market share.
10. Pressure to meet debt covenants or earnings expectations.

[LOS 1.f] Contrast absolute and relative valuation models and describe examples of each type of model.

	Absolute valuation models	Relative valuation models
Goal	Specifies an asset's intrinsic value.	Estimates an asset's value relative to that of another asset.
Model	Dividend discount model, free cash flow model, residual income model.	Multiples of market price
Underlying idea	The value of an asset is related to the returns expected to receive from holding that asset.	Similar assets should sell at similar prices.
Main approach	Determine the value of a firm today as the discounted or present value of all the cash flows expected in the future.	Using price multiples or enterprise multiples to compare to that of comparable firms .

[LOS 1.g] Describe sum-of-the-parts valuation and conglomerate discounts.

1.

Sum-of-the-parts valuation

Sum-of-the-parts valuation

Value individual parts of the firm and add them up to determine the value for the company as a whole.

This method is used to value company that operates in multiple divisions with different business models.



Company with multiple divisions:

- Have different characteristics (*eg: discount factors*) for each division;
- Lack of comparable companies.

**Difficulties in applying
Absolute/Relative valuation
models**

Solution: Sum-of-the-parts valuation

1. Valuation of individual parts



2. Add up for company value



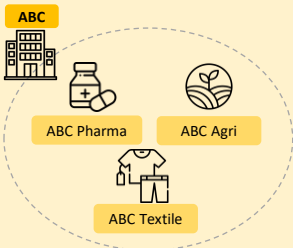
The value obtained is called the ***sum-of-the-parts value, breakup value or private market value.***

[LOS 1.g] Describe sum-of-the-parts valuation and conglomerate discounts.

2.

Conglomerate discount

Definition: **Conglomerate discount** is a markdown investors apply to the value of companies that operate in multiple unrelated industries.



ABC Group with **unrelated** business units (and no synergy)



Lower overall market value for the firm

Three explanations for conglomerate discounts

Internal capital inefficiency: Company can make poor choices allocating resources to its many units.

Endogenous (internal) factors: Pursued unrelated business for other purposes than making profit.

Research measurement errors: conglomerate discounts do not exist, just a result of incorrect measurement.

[LOS 1.h] Explain broad criteria for choosing an appropriate approach for valuing a given company.

A relevant valuation model for a company is a model that:

Fits the characteristics of the company.

For example, a company that does not pay dividend can not be valued with dividend discounting model.

Is appropriate based on the quality and availability of input data.

Is suitable given the purpose of the analysis.

For example, making a purchase offer for a controlling interest in the company allow the purchaser to set dividend policy. → A model based on cashflow is more appropriate than one based on dividend.

Note:

- Analyst does not have to consider only one model.
- Using multiple models and examining differences in estimated values can reveal how a model's assumptions and the perspective of the analysis are affecting the estimated values.

MODULE 2 : DISCOUNTED DIVIDEND VALUATION

LEARNING OUTCOMES

- 2.a.** Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and identify investment situations for which each measure is suitable
- 2.b.** Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and multiple holding periods
- 2.c.** Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions
- 2.d.** Calculate the value of non-callable fixed-rate perpetual preferred stock
- 2.e.** Describe strengths and limitations of the Gordon growth model and justify its selection to value a company's common shares
- 2.f.** Calculate and interpret the implied growth rate of dividends using the Gordon growth model and current stock price
- 2.g.** Calculate and interpret the present value of growth opportunities (PVGO) and the component of the leading price-to-earnings ratio (P/E) related to PVGO
- 2.h.** Calculate and interpret the justified leading and trailing P/Es using the Gordon growth model

LEARNING OUTCOMES

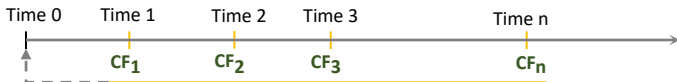
- 2.i.** Estimate a required return based on any DDM, including the Gordon growth model and the H-model
- 2.j.** Evaluate whether a stock is overvalued, fairly valued, or undervalued by the market based on a DDM estimate of value
- 2.k.** Explain the growth phase, transition phase, and maturity phase of a business
- 2.l.** Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or spreadsheet modeling to value a company's common shares
- 2.m.** Describe terminal value and explain alternative approaches to determining the terminal value in a DDM
- 2.n.** Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM
- 2.o.** Explain the use of spreadsheet modeling to forecast dividends and to value common shares
- 2.p.** Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis to estimate a company's sustainable growth rate

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and identify investment situations for which each measure is suitable

1.

Introduction: Discounted cash flow models

Discounted cash flow models are a fundamental tool which view the intrinsic value of common stock as the present value of its expected future cash flows.



Valuation based on Present Value of Future Cash Flows

$$V_0 = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \dots + \frac{CF_n}{(1+r)^n} = \sum_{t=1}^n \frac{CF_t}{(1+r)^t}$$

where:

V_0 is the value of a share of stock today, at $t = 0$

CF_t is the cash flow (or the expected cash flow, for risky cash flows) at time t

r is the discount rate or required rate of return

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

Example: Value as the Present Value of Future Cash Flows

An asset is expected to generate cash flows of \$100 in one year, \$150 in two years, and \$200 in three years.

The value of this asset today, using a 10% discount rate, is:

$$V_0 = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} = \frac{100}{1.1} + \frac{150}{1.1^2} + \frac{200}{1.1^3}$$

$$= 90.909 + 123.967 + 150.263 = \$365.14$$

In stock valuation models, there are **three** predominant definitions of future cash flows (CF)

Dividends

1.1

Free cash flow

1.2

Residual income

1.3

Throughout the discussion of Discounted cash flow model, we assume that cash flows for a period are paid in one sum **at the end** of the period.

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.1. Dividends:

Dividend discount models (DDMs) define cash flow as the dividends to be received by the shareholders.

CF = Dividends (D)

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_n}{(1+r)^n}$$

Advantage

Theoretically justified: The shareholder's investment today is worth the present value of the future cash flows expected to receive in the form of dividends.

Value estimates are less volatile and reflect the long-term earning potential of the company.

Disadvantage

Difficult to implement for firms that don't currently pay dividends and forecast the fundamental variables that influence stock.

Perspective of an investor who owns a minority stake in the firm and cannot control the dividend policy that may not also related to the firm's ability to create value.

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.1. Dividends:

 *In what cases should we use Dividends as a proxy for cash flow*

- The company is dividend-paying (i.e., the analyst has a dividend record to analyze)
 - The board of directors has established a dividend policy that bears an understandable and consistent relationship to the company's profitability
 - The investor takes a non-control perspective
- Most appropriate for seasoned companies that are profitable but operating outside the economy's fastest-growing subsectors.

In this module, we will delve into the details of Dividend discount models. Other discount models, such as Free cash flow and Residual income, which are illustrated in the next slides, will be discussed in detail in the upcoming modules.

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.2. Free cash flow:

Free cash flow discount models define cash flow as the free cash flow to the firm (FCFF) or free cash flow to equity (FCFE).

FCFF

Definition

Cash flow generated by the firm's operations that is in excess of the capital investment required to sustain the firm's current productive capacity.

FCFE

Cash available to stockholders after funding capital requirements and expenses associated with debt financing.

CF = Free cash flow (FCF)

Equity Valuation

$$V_0 = \frac{FCFE_1}{(1+r)^1} + \frac{FCFE_2}{(1+r)^2} + \dots + \frac{FCFE_n}{(1+r)^n}$$

$$V_0 = \frac{FCFF_1}{(1+WACC)^1} + \frac{FCFF_2}{(1+WACC)^2} + \dots + \frac{FCFF_n}{(1+WACC)^n} - \text{MV}_{\text{Outstanding debt}}$$

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.2. Free cash flow:

Advantage

Can be applied to many firms, regardless of dividend policies or capital structures.

Valuable for both controlling and minority shareholders because they allow control over cash flow distribution and the potential for market value acquisition.

Disadvantage

Firms that have significant capital requirements may have negative free cash flow for many years into the future → Makes the estimates less reliable.

Changes in interest rates or borrowing costs can significantly impact the present value of free cash flows → Makes the estimates less reliable.

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.2. Free cash flow:



In what cases should we use Free cash flow as a proxy for cash flow

- The company is not dividend-paying
- The company is dividend-paying but dividends significantly exceed or fall short of free cash flow to equity
- The company's free cash flows align with the company's profitability within a forecast horizon with which the analyst is comfortable
- The investor takes a control perspective

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.3. Residual income:

Residual income discount models define cash flow as the residual income, which is the amount of earnings during the period that exceeds the investors' required return.

CF = Residual income (RI)

$$V_0 = B_0 + \frac{RI_1}{(1+r)^1} + \frac{RI_2}{(1+r)^2} + \dots + \frac{RI_n}{(1+r)^n}$$

Advantage

Can be applied to firms with negative free cash flow and to dividend- and non-dividend-paying firms.

Less sensitive to short-term fluctuations in earnings → Makes it a valuable tool for assessing the long-term value of a company.

Disadvantage

More difficult to apply, because they require in-depth analysis of the firm's accounting accruals.

The accounting may lack transparency or the quality of the firm's reporting may be poor → Makes the estimates less reliable.

[LOS 2.a] Compare dividends, free cash flow, and residual income as inputs to discounted cash flow models and...

1.

Introduction: Discounted cash flow models

1.3. Residual income:



In what cases should we use Residual income as a proxy for cash flow

- The company is not paying dividends, as an alternative to a free cash flow model
- The company's expected free cash flows are negative within the analyst's comfortable forecast horizon
- The company with transparent financial reporting and high-quality earnings

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and multiple holding periods

1.

The Expression for a Single Holding Period

If an investor wishes to buy a share of stock and hold it for **one year**, the value of that share of stock today is the present value of the **expected dividend** to be received on the stock plus the present value of the **expected selling price** in one year.



One-Period DDM

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{P_1}{(1+r)^1} = \frac{D_1 + P_1}{1+r}$$

where:

V_0 is the value of a share of stock today, at $t = 0$

P_1 is the expected price per share at $t = 1$

D_1 is the expected dividend per share for Year 1

r is the required rate of return on the stock

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

1.

The Expression for a Single Holding Period

Example: DDM Value with a Single Holding Period

Suppose that you expect Carrefour SA (CA: EN Paris) to pay a €0.46 dividend next year. You expect the price of Carrefour stock to be €23.00 in one year. The required rate of return for Carrefour stock is 8%. What is your estimate of the value of Carrefour stock?

Answer:

Discounting the expected dividend of €0.46 ($D_1 = €0.46$) and the expected sales price of €23.00 ($P_1 = €23.00$) at the required return on equity of 8% ($r = 8\%$), we obtain:

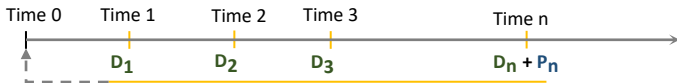
$$V_0 = \frac{D_1 + P_1}{1 + r} = \frac{0.46 + 23.00}{1 + 0.08} = 21.72$$

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

2.

The Expression for Multiple Holding Periods

For an n-period model, the value of a stock is the present value of the **expected dividends** for the n periods plus the present value of the **expected price** in n periods (at t = n).



Multi-Period DDM

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_n + P_n}{(1+r)^n}$$

where:

V_0 is the value of a share of stock today, at t = 0

P_n is the expected price per share at t = n

D_i is the expected dividend per share for period i ($\forall i=1,n$)

r is the required rate of return on the stock

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

2.

The Expression for Multiple Holding Periods

Example 1: Finding the Stock Price for a Five-Year Forecast Horizon

For the next five years, the annual dividends of a stock are expected to be \$2.00, \$2.10, \$2.20, \$3.50, and \$3.75. In addition, the stock price is expected to be \$40.00 in five years. If the required return on equity is 10%, what is the value of this stock?

Answer:

We have the following table:

D_1	D_2	D_3	D_4	D_5	P_5
\$2.00	\$2.10	\$2.20	\$3.50	\$3.75	\$40.00

The present values of the expected future cash flows can be written out as:

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_5 + P_5}{(1+r)^5}$$

Calculating and summing these present values gives a stock value of $V_0 = 1.818 + 1.736 + 1.653 + 2.391 + 2.328 + 24.837 = \34.76

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

2.

The Expression for Multiple Holding Periods

Example 2: Calculating value for a three-period DDM

Reliable Motors shares are expected to pay dividends of \$1.50, \$1.60, and \$1.75 at the end of each of the next three years, respectively. The investor expects the price of the shares at the end of this 3-year holding period to be \$54.00. The investor's required rate of return is 15%. Calculate the current value of Reliable's shares.

Answer:

We have the following table:

D_1	D_2	D_3	P_3
\$1.50	\$1.60	\$1.75	\$54.00

When we have to calculate the total of three or more discounted cash flows, we can generally save a considerable amount of time (and improve accuracy) by using our financial calculators as shown in the next slide.

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

2.

The Expression for Multiple Holding Periods

Example 2: Calculating value for a three-period DDM

Calculating the PV of Multiple Cash Flows With the TI BA II Plus™

Key Strokes	Explanation	Display
[CF] → [2 ND] → [CLR WORK]	Clear memory registers	CF ₀ = 0
[ENTER]	Initial cash outlay	CF ₀ = 0
[↓] → 1.50 → [ENTER]	Year 1 cash flow	C01 = 1.50
[↓]	Frequency of cash flow 1	F01 = 1
[↓] → 1.60 → [ENTER]	Year 2 cash flow	C02 = 1.60
[↓]	Frequency of cash flow 2	F02 = 1
[↓] → 55.75 → [ENTER]	Year 3 cash flow	C03 = 55.75
[↓]	Frequency of cash flow 3	F03 = 1
[↓][NPV] → 15 → [ENTER]	15% discount rate	I = 15
[↓] → [CPT]	Calculate NPV of all CFs	NPV = 39.17



[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

3.

Warm-Up: The General Dividend Discount Model

If we extend the holding period **indefinitely**, the value simply becomes the present value of an **infinite stream of dividends**, represented by John Burr Williams's (1938) original DDM formula.

Infinite-Period DDM

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

To use the DDM, the forecasting problem must be simplified. **Two** broad approaches exist, each of which has several variations:

01

Future dividends can be forecast by assigning the stream of future dividends to one of several stylized growth patterns.

The most commonly used patterns are:

- Constant growth forever (the Gordon growth model)
- Two distinct stages of growth (the two-stage growth model and the H-model)
- Three distinct stages of growth (the three-stage growth model)

We will present the Gordon growth model, the two stage H-model, and three-stage growth models later.

[LOS 2.b] Calculate and interpret the value of a common stock using the dividend discount model (DDM) for single and...

3.

Warm-Up: The General Dividend Discount Model

To use the DDM, the forecasting problem must be simplified. **Two** broad approaches exist, each of which has several variations:

02

A finite number of dividends can be forecast individually up to a terminal point, typically from 3 to 10 years into the future.

The horizon selected often depends on the perceived predictability (sometimes called the **visibility**) of the company's earnings.

We can then forecast either:

- The remaining dividends from the terminal point forward by assigning
- The share price at the terminal point of our dividend forecasts (terminal share price)

Spreadsheets are particularly convenient tools for implementing a DDM with individual dividend forecasts but are useful in all cases. We will address spreadsheet modeling at a later stage.

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

The **Gordon growth model (GGM)** developed by Gordon and Shapiro (1956) and Gordon (1962), assumes that dividends **grow indefinitely** at a **constant rate**.

Infinite-Period DDM

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$



$$D_t = D_{t-1}(1+g) = D_0(1+g)^t$$

Infinite-Period GGM

$$V_0 = \frac{D_0(1+g)}{(1+r)^1} + \frac{D_0(1+g)^2}{(1+r)^2} + \dots + \frac{D_0(1+g)^n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t}$$

where:

D_0 is the dividend just paid

r is the required rate of return on the stock

g is the dividend growth rate

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

Infinite-Period GGM

$$V_0 = \frac{D_0(1+g)}{(1+r)^1} + \frac{D_0(1+g)^2}{(1+r)^2} + \dots + \frac{D_0(1+g)^n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t}$$

We have **three** underlying assumptions of GGM as below:

01 The firm expects to pay a dividend, D_1 , in one year.

Note: The dividend to be discounted is the next period dividend, D_1 , not the dividend from the previous period, D_0 .

02 Dividends grow indefinitely at a constant rate, g (which may be less than zero).

03 The growth rate, g , is less than the required return, r .

Note: This is very important assumption to simplify the GGM equation above (see next slide).

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

Infinite-Period GGM

$$V_0 = \frac{D_0(1+g)}{(1+r)^1} + \frac{D_0(1+g)^2}{(1+r)^2} + \dots + \frac{D_0(1+g)^n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t}$$



How can we simplify the equation above

Mathematical knowledge: Geometric progression

A geometric progression is a sequence of numbers in which each term after the first is found by multiplying the previous term by a fixed, non-zero number.

A geometric progression typically takes the following form:

$$a, am, am^2, am^3, \dots$$

With $|m| < 1$, we have the sum of an infinite geometric progression:

$$a + am + am^2 + am^3 + \dots = \frac{a}{1-m} (1)$$

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

Infinite-Period GGM

$$V_0 = \frac{D_0(1+g)}{(1+r)^1} + \frac{D_0(1+g)^2}{(1+r)^2} + \dots + \frac{D_0(1+g)^n}{(1+r)^n} + \dots = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t}$$

 How can we simplify the equation above

With $|m| < 1$, we have the sum of an infinite geometric progression:

$$a + am + am^2 + am^3 + \dots = \frac{a}{1-m} \quad (1)$$

From equation (1), setting $a = D_0$ and $m = \frac{1+g}{1+r}$ ($|m| < 1$ due to assumption 3 of GGM that $g < r$) we get the Gordon growth model and its simplification.

GGM's simplification

$$V_0 = \frac{D_0(1+g)}{r-g} \text{ or } V_0 = \frac{D_1}{r-g}$$

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

- A firm's growth rate projections can be compared to the growth rate of the economy to determine if it can continue indefinitely.
- It is unrealistic to assume that any firm can continue to grow indefinitely at a rate higher than the long-term growth rate in real gross domestic product (GDP) plus the long-term inflation rate.
- In general, a perpetual dividend growth rate forecast above 5% is suspect.

[LOS 2.c] Calculate the value of a common stock using the Gordon growth model and explain the model's underlying assumptions

1.

The Gordon Growth Model

Example: Calculating value with the Gordon growth model

DownUnder Financial recently paid a dividend of 1.80 Australian dollars (A\$). An analyst has examined the financial statements and historical dividend policy of DownUnder and expects that the firm's dividend rate will grow at a constant rate of 3.5% indefinitely. The analyst also determines DownUnder's beta is 1.5, the riskfree rate is 4%, and the expected return on the market portfolio is 8%. Calculate the current value of DownUnder's shares.

Answer:

Applying the GGM's simplification equation we have:

$$V_0 = \frac{D_0(1+g)}{r-g}$$

We are given $D_0 = \text{A\$}1.80$, $g = 3.5\%$. Now we need to calculate r by using the capital asset pricing model (CAPM) as the following:

$r = \text{riskfree rate} + \text{beta}(\text{expected return on the market portfolio} - \text{riskfree rate}) = 4\% + 1.5(8\% - 4\%) = 10\%$.

Thus, $V_0 = \frac{\text{A\$}1.8(1 + 0.035)}{0.1 - 0.035} = \text{A\$}28.66$

[LOS 2.d] Calculate the value of non-callable fixed-rate perpetual preferred stock

1.

Fixed-rate perpetual preferred stock

Fixed-rate perpetual preferred stock is the stock with a specified dividend rate that has a claim on earnings senior to the claim of common stock, and no maturity date

Perpetual preferred stock has been used particularly by financial institutions such as banks to obtain permanent equity capital while diluting the interests of common equity.

Valuation of the non-callable form of fixed-rate perpetual preferred stock is **straightforward**: The **Gordon growth model** can also be used as the following:

Perpetual preferred shares' valuation

$$V_0 = \frac{D_p}{r_p}$$

where:

D_p is preferred dividend (which is assume not to grow ($g = 0$))

r_p is cost of preferred equity (also called capitalization rate)

[LOS 2.d] Calculate the value of non-callable fixed-rate perpetual preferred stock

1.

Fixed-rate perpetual preferred stock

Example: Calculating the value of fixed-rate perpetual preferred stock

United Publishing has a fixed-rate perpetual preferred stock outstanding with a dividend of 6% (based on an issue at par of £100). If the investors' required rate of return for holding these shares is 9.5%, calculate the current value of these shares.

Answer:

Applying the Perpetual preferred shares' valuation equation we have:

$$V_0 = \frac{D_p}{r_p}$$

We are given $r_p = 9.5\%$. Now we need to calculate D_p .

Dividends are not growing because the preferred dividend is based on a fixed rate of 6.0% on a stated par value of £100, so this is comparable to DDM with a fixed dividend in perpetuity, and:

$$D_p = 0.06 \times £100 = £6.00$$

$$\text{Thus, } V_0 = \frac{£6.00}{0.095} = £63.16$$

[LOS 2.e] Describe strengths and limitations of the Gordon growth model and justify its selection to value a company's common shares

1.

Strengths and limitations of the Gordon growth model

The Gordon growth model (GGM) has a number of characteristics that make it useful and appropriate for many applications, however, there are also some characteristics that limit the applications of the GGM as below:

Strengths

Applicable to stable, mature, dividend-paying firms.

Appropriate for valuing market indices

Easily communicated and explained because of its straightforward approach

Can be used to determine price-implied growth rates, required rates of return, and value of growth opportunities

Can be used to supplement other, more complex valuation methods

Limitations

Valuations are very sensitive to estimates of growth rates and required rates of return, both of which are difficult to estimate with precision.

The model cannot be easily applied to non-dividend-paying stocks

Unpredictable growth patterns of some firms would make using the model difficult and the resulting valuations unreliable.

[LOS 2.e] Describe strengths and limitations of the Gordon growth model and justify its selection to value a company's common shares

2.

Value Appreciation in the Gordon Growth Model

The Gordon growth model implies a set of relationships for the growth rates of dividends, earnings, and stock value

$$V_0 = \frac{D_1}{r - g} \quad (1) \rightarrow V_1 = \frac{D_2}{r - g}$$

→ Both **dividends**, **value** and **earning** have grown at a rate of g (holding r and payout ratio $(\frac{D}{E})$ constant)

→ g is the rate of value or capital appreciation (sometimes also called the capital gains yield)

Dividend yield: $\frac{D_1}{P_0} = r - g$ (from equation (1)) → **Dividend yield is constant.**

(also because both the dividend and the price are expected to grow at the same rate g)

Capital gains yield: g (as discussed above) → **Capital gains yield is constant.**

Total return = Dividend yield + Capital gains yield = $r - g + g = r$ will also stay constant through time, given that price tracks value exactly.

[LOS 2.f] Calculate and interpret the implied growth rate of dividends using the Gordon growth model and current stock price

1.

Share repurchases

Share repurchases involve a reduction in the number of shares outstanding, all else equal. They are generally harder to forecast than the cash dividends of companies with an identifiable dividend policy.

*When companies have share repurchase programs, an analyst has **two** method to calculate the valuation:*

01

Focus on total payout (i.e., dividends plus repurchases) and use that in their valuation model.

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t + SR_t}{(1+r)^t}$$

where SR_t is repurchases in time t

02

Focus on cash dividends discount model after adjusting for the number of shares outstanding.

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

where g and r are now taken into account for share repurchases

Correctly applied, the DDM is a valid approach to common stock valuation even when the company being analyzed engages in share repurchases.

[LOS 2.f] Calculate and interpret the implied growth rate of dividends using the Gordon growth model and current stock price

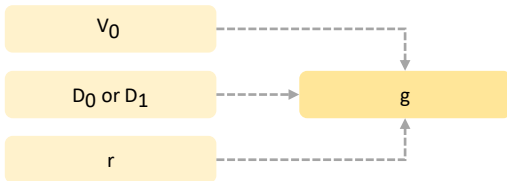
2.

The Implied Dividend Growth Rate

The Gordon growth model includes four variables, so if we know any three of them, we can solve for the fourth. Here, we calculate the implied growth rate of dividends (g).

GGM's simplification

$$V_0 = \frac{D_0(1+g)}{r-g} \text{ or } V_0 = \frac{D_1}{r-g}$$



[LOS 2.f] Calculate and interpret the implied growth rate of dividends using the Gordon growth model and current stock price

2.

The Implied Dividend Growth Rate

Example: Calculating the implied growth rate using the Gordon growth model

Suppose that the current price and most recent annual dividend for Aurora Mining (AM) are \$24.25 and \$1.10, respectively. If the required return on Aurora is 8.5%, what is the implied growth rate?

Answer:

We can set up the Gordon growth model in its standard form using the information we know:

$$V_0 = \frac{D_0(1+g)}{r-g}$$

We are given $V_0 = \$24.25$, $D_0 = \$1.10$ and $r = 8.5\%$. Thus, the equation above becomes: $24.25 = \frac{1.10(1+g)}{0.085-g}$.

Rearranging the terms gives us:

$$1.10 + 1.10 \times g = 2.06125 - 24.25 \times g$$

$$\rightarrow g = 0.0379$$

So, assuming that our estimated required return is on target, the implied growth rate for Aurora Mining's dividends is 3.8%.

[LOS 2.g] Calculate and interpret the present value of growth opportunities (PVGO) and the component of the leading price-to-earnings ratio (P/E) related to PVGO

1.

Present value of growth opportunities (PVGO)

Value of a stock (V_0)

Value of the company without earnings reinvestment (V_1)

1.1

Present value of growth opportunities (PVGO)

1.2

1.1. Value of the company without earnings reinvestment:

Increases in shareholder wealth occur only when reinvested earnings earn more than the opportunity cost of funds ($ROE > r$) - that is, when investments are in **positive NPV** projects.

A company without positive expected NPV projects is defined as a **no growth company** ($g = 0$).

The company should distribute all their earnings in dividends ($D_1 = E_1$).

[LOS 2.g] Calculate and interpret the present value of growth opportunities (PVGO) and the component of...

1.

Present value of growth opportunities (PVGO)

The company should distribute all their earnings in dividends.

No growth value per share

$$V_1 = \frac{E_1}{r}$$

1.2. Present value of growth opportunities (PVGO):

PVGO (value of growth), which sums the expected value today of opportunities to profitably reinvest future earnings, can be defined as the forecasted total net present value of future projects.

If prices reflect value ($P_0 = V_0$)

Value of growth

$$PVGO = V_0 - V_1 = P_0 - \frac{E_1}{r}$$

[LOS 2.g] Calculate and interpret the present value of growth opportunities (PVGO) and the component of...

2.

The component of the leading P/E related to PVGO

As discussed, for any company, the actual value per share is the sum of the no growth value per share and the present value of growth opportunities:

Value of a stock

$$V_0 = V_1 + \text{PVGO} = \frac{E_1}{r} + \text{PVGO}$$

Leading price-to-earnings ratio

$$\frac{V_0}{E_1} \text{ or } \frac{P_0}{E_1} \text{ or } P/E = \frac{1}{r} + \frac{\text{PVGO}}{E_1}$$

$$\frac{1}{r}$$

$$\frac{\text{PVGO}}{E_1}$$

The value of the P/E for a no growth company

The component of the P/E value that relates to growth opportunities

[LOS 2.g] Calculate and interpret the present value of growth opportunities (PVGO) and the component of...

2.

The component of the leading P/E related to PVGO

Example: Calculating PVGO

Reliable, Inc.'s shares trade at 60.00 Swiss francs (Sf) with expected earnings of Sf5.00 per share and a required return of 10%. Suppose that the shares are properly priced, so price is equal to fundamental value. Calculate the PVGO and the portion of the leading P/E related to PVGO.

Answer:

According to Value of growth's equation, we have:

$$PVGO = P_0 - \frac{E_1}{r}$$

We are given $P_0 = \text{Sf}60.00$, $E_1 = \text{Sf}5.00$ and $r = 10\%$.

$$\text{Thus, } PVGO = 60.00 - \frac{\text{Sf}5.00}{0.1} = \text{Sf}10.00$$

The portion of the leading P/E related to PVGO is:

$$\frac{\frac{PVGO}{E_1}}{\frac{1}{r} + \frac{PVGO}{E_1}} = \frac{\frac{10.00}{5.00}}{\frac{1}{0.1} + \frac{10.00}{5.00}} = 16.7\%$$

Thus, 16.7% of the firm's leading P/E ratio is attributable to PVGO.

[LOS 2.h] Calculate and interpret the justified leading and trailing P/E's using the Gordon growth model

1.

Justified P/E's using the Gordon growth model

The price-to-earnings (P/E) ratio is the most commonly used relative valuation indicator. A **justified P/E** is derived based on the firm's fundamentals.

*There are **two** most common form of justified P/E. Both of these can be developed from the Gordon growth model:*

Justified leading P/E

It is based on the earnings forecast for the next period (E_1)

Justified leading P/E

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r - g} = \frac{1 - b}{r - g}$$

Justified trailing P/E

It is based on the earnings for the previous period (E_0)

Justified trailing P/E

$$\frac{P_0}{E_0} = \frac{D_0(1 + g)/E_0}{r - g} = \frac{(1 - b)(1 + g)}{r - g}$$

where:

P_0 is fundamental value

b is retention ratio

$1 - b$ is dividend payout ratio

[LOS 2.h] Calculate and interpret the justified leading and trailing P/Es using the Gordon growth model

1.

Justified P/Es using the Gordon growth model

Example: Calculating justified leading and trailing P/E

Alliance, Inc., is currently selling for \$16.00 on current earnings of \$3.00 and a current dividend of \$1.50. Dividends are expected to grow at 3.5% per year indefinitely. The risk-free rate is 4%, the market equity risk premium is 6%, and Alliance's beta is estimated to be 1.1. Calculate the justified leading and trailing P/E ratios of Alliance, Inc.

Answer:

- Using the capital asset pricing model (CAPM):

Required return (r) = risk-free rate + beta $\times$ market equity risk premium
 $= 4\% + 1.1 \times 6\% = 10.6\%$.

- Retention ratio (b) = (earnings – dividends) / earnings
 $= (\$3.00 - \$1.50) / \$3.00 = 0.5$

We are given $g = 3.5\%$. Thus, we have:

- Justified leading P/E:** $\frac{P_0}{E_1} = \frac{1 - b}{r - g} = \frac{1 - 0.5}{0.106 - 0.035} = 7.04$
- Justified trailing P/E:** $\frac{P_0}{E_0} = \frac{(1 - b)(1 + g)}{r - g} = \frac{(1 - 0.5)(1 + 0.035)}{0.106 - 0.035} = 7.29$

or **Justified trailing P/E = Justified leading P/E $\times$ (1 + g)**

$= 7.04 \times (1 + 0.035) = 7.29$.

[LOS 2.k] Explain the growth phase, transition phase, and maturity phase of a business

1.

Three phase of a business

While the basic GGM assumes constant growth, most firms go through a pattern of growth that includes **three** phases as the following:

Initial growth phase

The firm has rapidly increasing earnings, little or no dividends, and heavy reinvestment.

Transition phase

Earnings and dividends are still increasing but at a slower rate as competitive forces reduce profit opportunities and the need for reinvestment.

Mature phase

Earnings grow at a stable but slower rate, and payout ratios are stabilizing as reinvestment matches depreciation and asset maintenance requirements.

*The level and pattern of specific fundamental variables during the **three** phases, as well as the appropriate valuation model to apply in each phase, are shown in the next slide.*

[LOS 2.k] Explain the growth phase, transition phase, and maturity phase of a business

1.

Three phase of a business

Variable	Growth phase		
	Initial growth	Transition	Maturity
Earnings growth	Very high	Above average but falling	Stable at long-run level
Capital investment	Significant requirements	Decreasing	Stable at long-run level
Profit margin	High	Above average but falling	Stable at long-run level
FCFE	Negative	Maybe positive and growing	Stable at long-run level
ROE & required return	$ROE > r$	ROE approaching r	$ROE = r$
Dividend payout	Low or zero	Increasing	Stable at long-run level
Appropriate model	Three-stage model (<i>discuss later</i>)	<i>Two-stage model (discuss later)</i>	Gordon growth model

[LOS 2.I] Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or spreadsheet modeling to value a company's common shares

1.

Multistage Dividend Discount Models

For most companies, the **Gordon growth model assumption** of constant dividend growth that continues into perpetuity is **unrealistic**.

We need more realistic **multistage growth models** to estimate value for companies with several stages of future growth.

Two-stage DDM

H-model

Three-stage model

Spreadsheet modeling

Whichever multistage model we use, there are three important points to keep in mind:

- We're still just forecasting dividends into the future and discounting them back to today to find intrinsic value.
- Over the long term, growth rates tend to revert to a long-run rate approximately equal to the long-term growth rate in real gross domestic product (GDP) plus the long-term inflation rate.
- The required rate of return applicable to each stage might be different.

[LOS 2.I] Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or...

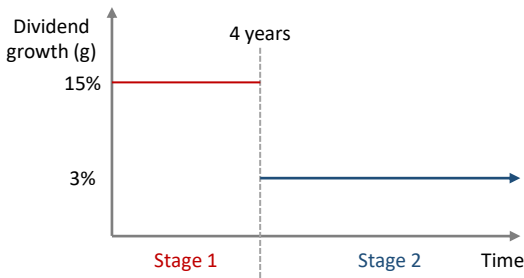
2.

Two-Stage DDM

Two-Stage DDM is the most basic multistage model in which we assume the company grows at a high rate for a relatively short period of time (the first stage) and then reverts to a long-run perpetual growth rate (the second stage).

Example of a Two-Stage DDM:

Suppose a firm is expected to grow at 15% until patents expire in four years, then immediately revert to a long-run growth rate of 3% in perpetuity.



This stock should be modeled by a two-stage model, with dividends growing at 15% before the patent expires and 3% thereafter.

[LOS 2.I] Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or...

3.

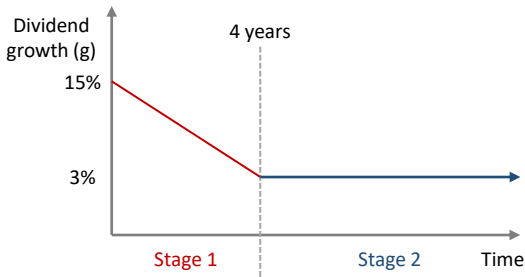
H-model

The problem with the **basic two-stage DDM**: usually unrealistic to assume that a stock will experience high growth for a short period, then immediately fall back to a long-run level.

H-model is a version of Two-stage DDM in which the dividend growth rate is assumed to decline from an abnormal rate to the mature growth rate during the course of Stage 1.

Example of an H-model:

Consider a firm that is currently growing at 15% and its growth rate will decline by 3% per year until it reaches 3% at the end of the fourth year.



[LOS 2.I] Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or...

3.

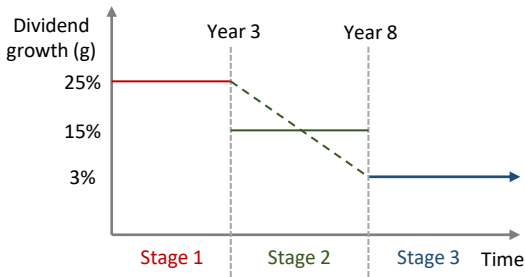
Three-stage DDM

Three-Stage DDM is appropriate for firms that are expected to have three distinct stages of earnings growth. It is a slightly more complex refinement of a two-stage model.

Example of a Three-Stage DDM:

Suppose we forecast that a biotech company will experience supernormal growth of 25% for three years, then 15% growth for five years, and finally slow down to a stable, long-run rate of 3%.

Alternatively, in Stage 2, the growth rate may also linearly decay to the Stage 3 stable, long-run growth rate.



[LOS 2.I] Explain the assumptions and justify the selection of the two-stage DDM, the H-model, the three-stage DDM, or...

4. Spreadsheet modeling

In practice we can use **spreadsheets** to model **any pattern of dividend growth** we'd like with different growth rates for each year because the spreadsheet does all the calculations.

Spreadsheet modeling is applicable to firms about which we have a great deal of information and can project different growth rates for differing periods.

Example of three different Spreadsheet models:

Growth Rates

Year	1	2	3	4	5	6	7 and after
Scenario 1	20%	19%	13%	5%	5%	5%	5%
Scenario 2	20%	19%	13%	11%	5%	5%	5%
Scenario 3	20%	19%	13%	11%	8%	7%	5%

[LOS 2.m] Describe terminal value and explain alternative approaches to determining the terminal value in a DDM

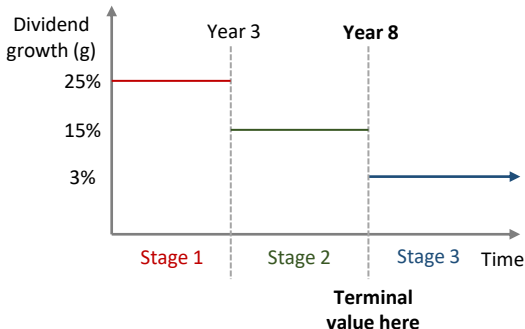
1.

Terminal value

Terminal value is the estimated value of an investment at a specific point in the future. It represents the present value of all expected future cash flows beyond a certain time horizon, usually when the firm is assumed to grow at a stable and sustainable rate.

No matter which dividend discount model we use, we have to estimate a terminal value at some point in the future.

Example of a Three-Stage DDM:



[LOS 2.m] Describe terminal value and explain alternative approaches to determining the terminal value in a DDM

2.

Alternative approaches to determining the terminal value

There are **two** ways to estimate the terminal value:

Using the Gordon growth model

At some point in the future, we assume dividends will begin to grow at a constant, long-term rate. → The terminal value at that point is just the value derived from the Gordon growth model.

Terminal value using GGM

$$V_t = \frac{D_t(1+g)}{r-g} \text{ or } V_t = \frac{D_{t+1}}{r-g}$$

Using the market multiple approach

We use market price multiples, especially a P/E ratio at the forecast horizon and then estimate the terminal value as the P/E multiplied by the earnings estimate.

Terminal value using MM

$$V_t = \frac{P_t}{E_t} \times E_t \text{ or } V_t = \frac{P_t}{E_{t+1}} \times E_{t+1}$$

[LOS 2.m] Describe terminal value and explain alternative approaches to determining the terminal value in a DDM

2.

Alternative approaches to determining the terminal value

Example: Estimating terminal value

Level Partners is expected to have earnings in ten years of \$12 per share, a dividend payout ratio of 50%, and a required return of 11%. At that time, the dividend growth rate is expected to fall to 4% in perpetuity, and the trailing P/E ratio is forecasted to be eight times earnings. Estimate the terminal value at the end of ten years using the Gordon growth model and the P/E multiple.

Answer:

Recall that we have officially denoted “b” as retention rate and “1 – b” as dividend payout ratio.

The dividend at the end of ten years is: $D_{10} = E_{10} \times (1 - b) = \$12 \times 50\% = \$6$.

The terminal value at the end of ten years using the Gordon growth model

$$\text{is: } V_{10} = \frac{D_{10}(1 + g)}{r - g} = \frac{6 \times (1 + 0.04)}{0.11 - 0.04} = \$89.14.$$

The terminal value at the end of ten years using the P/E multiple is:

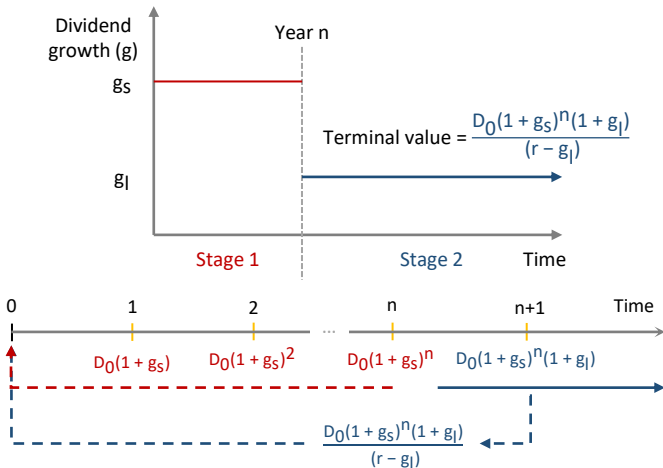
$$V_{10} = \frac{P_{10}}{E_{10}} \times E_{10} = 8 \times \$12 = \$96.$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

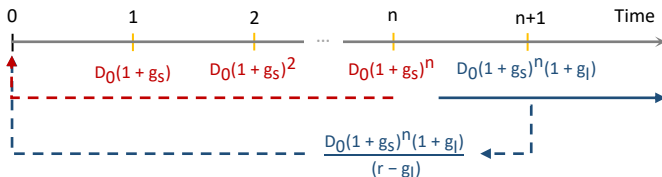
1.

The value of common shares using the two-stage DDM

Recall: The **two-stage fixed growth rate model** is based on the assumption that the firm will enjoy an initial period of high growth (g_S), followed by a mature or stable period in which growth will be lower but sustainable (g_L).



[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

1.
The value of common shares using the two-stage DDM


Two-stage DDM Valuation

$$V_0 = V_{S1} + V_{S2} = \sum_{t=1}^n \frac{D_0(1+g_S)^t}{(1+r)^t} + \frac{D_0(1+g_S)^n(1+g_L)}{(1+r)^n(r-g_L)}$$

where:

g_S is short-term growth rate

g_L is long-term growth rate

r is required return

n is length of high growth period

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

1.

The value of common shares using the two-stage DDM

Example: Calculating value with a two-stage DDM

Sea Island Recreation currently pays a dividend of \$1.00. An analyst forecasts growth of 10% for the next three years, followed by 4% growth in perpetuity thereafter. The required return is 12%. Calculate the current value per share.

Answer:

We could solve the problem by plugging the appropriate numbers into the formula as follows:

$$\begin{aligned}
 V_0 &= \sum_{t=1}^3 \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{D_0(1+g_s)^3(1+g_l)}{(1+r)^3(r-g_l)} \\
 &= \sum_{t=1}^3 \frac{\$1.00 \times (1.10)^t}{(1.12)^t} + \frac{\$1.00 \times (1.10)^3 \times (1.04)}{(1.12)^3 \times (0.12 - 0.04)} \\
 &= \$15.21
 \end{aligned}$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

1.

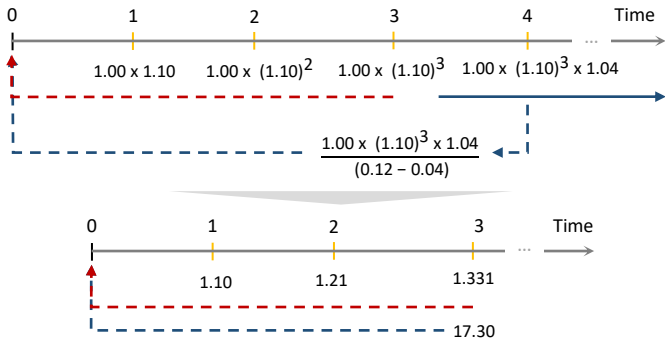
The value of common shares using the two-stage DDM

Example: Calculating value with a two-stage DDM

Sea Island Recreation currently pays a dividend of \$1.00. An analyst forecasts growth of 10% for the next three years, followed by 4% growth in perpetuity thereafter. The required return is 12%. Calculate the current value per share.

Answer:

When using calculator, we need to forecast dividend cash flows as the following:



[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

1.
The value of common shares using the two-stage DDM
Example: Calculating value with a two-stage DDM
Answer:

The financial calculator does the hard work for us:

Key Strokes	Explanation	Display
[CF] → [2 ND] → [CLR WORK]	Clear memory registers	CF ₀ = 0
[ENTER]	Initial cash outlay	CF ₀ = 0
[↓] → 1.10 → [ENTER]	Year 1 cash flow	C01 = 1.10
[↓]	Frequency of cash flow 1	F01 = 1
[↓] → 1.21 → [ENTER]	Year 2 cash flow	C02 = 1.21
[↓]	Frequency of cash flow 2	F02 = 1
[↓] → 18.63 → [ENTER]	Year 3 cash flow	C03 = 18.63
[↓]	Frequency of cash flow 3	F03 = 1
[↓][NPV] → 12 → [ENTER]	12% discount rate	I = 12
[↓] → [CPT]	Calculate NPV of all CFs	NPV = 15.21



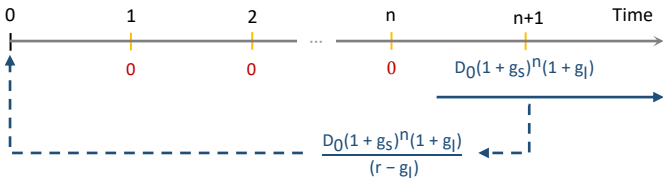
[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

2.

Valuing a Non-Dividend-Paying Company

The value of a firm that doesn't currently pay a dividend is a simple version of the two-stage DDM, where the firm pays no dividends in the first stage.

→ The value of the firm is just the present value of the terminal value computed at the point in time at which dividends are projected to start.



Two-stage Non-dividend paying Valuation

$$V_0 = V_{S2} = \frac{D_0(1 + g_S)^n(1 + g_I)}{(1 + r)^n(1 + g_I)}$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

2.

Valuing a Non-Dividend-Paying Company

Example: Valuing a non-dividend-paying stock

Arena Distributors is a new company and currently pays no dividends. The company recently reported earnings of \$1.50 per share and is expected to grow at a 15% rate for the next four years. Beginning in Year 5, Arena is expected to distribute 20% of its earnings in the form of dividends and to have a constant growth rate of 5%. The required rate of return is 12%. Calculate the value of Arena shares today.

Answer:

First, we need to calculate then terminal value in Year 4: $V_4 = \frac{D_5}{r - g_1}$

Recall that we have officially denoted “b” as retention rate and “1 – b” as dividend payout ratio.

$$\rightarrow D_5 = E_5 \times (1 - b) \text{ and we have: } E_5 = E_0 \times (1 + g_S)^4 \times (1 + g_I)$$

$$\rightarrow D_5 = E_0 \times (1 + g_S)^4 \times (1 + g_I) \times (1 - b) = \$1.50 \times 1.15^4 \times 1.05 \times 0.20 = \$0.55$$

$$\rightarrow V_4 = \frac{D_5}{r - g_1} = \frac{\$0.55}{0.12 - 0.05} = \$7.86$$

Thus, the value of Arena shares today is:

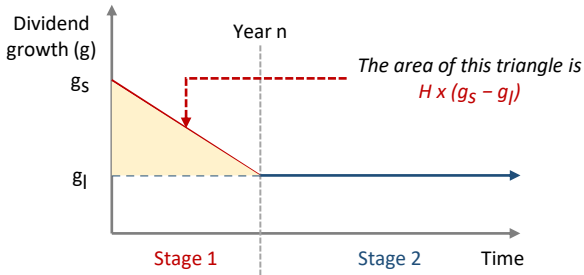
$$V_0 = \frac{V_4}{(1 + r)^4} = \frac{\$7.86}{1.12^4} = \$5.00.$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

3.

The value of common shares using the H-model

Recall: The H-model approximates the value of a firm assuming that an initially high rate of growth declines linearly over a specified period.



H-model Valuation

$$V_0 = V_{S12} + V_{S1} = \frac{D_0(1 + g_l)}{r - g_l} + \frac{D_0 \times H \times (g_s - g_l)}{r - g_l}$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

3.

The value of common shares using the H-model

where:

V_{S12} is the present value of the company's dividend stream if it were to grow at g_l forever

V_{S1} is an approximation of the extra value (assuming $g_s > g_l$) accruing to the stock because of its supernormal growth for Years 1 through 2H

g_s is short-term growth rate

g_l is long-term growth rate

r is required return

$H = \frac{n}{2}$ is half-life in years of the high-growth period.

Note: The H-model provides only an approximation of the value of shares.

$g_s - g_l \downarrow$

And/
or

$n \text{ or } H \downarrow$

H-model valuation's accuracy $\uparrow$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

3.

The value of common shares using the H-model

Example: Calculating value with the H-model

Omega Foods currently pays a dividend of €2.00. The growth rate, which is currently 20%, is expected to decline linearly over the next ten years to a stable rate of 5% thereafter. The required return is 12%. Calculate the current value of Omega.

Answer:

Applying the H-model valuation equation we have:

$$\begin{aligned}
 V_0 &= V_{S12} + V_{S1} = \frac{D_0(1 + g_l)}{r - g_l} + \frac{D_0 \times H \times (g_s - g_l)}{r - g_l} \\
 &= \frac{€2.00 \times (1 + 0.05)}{0.12 - 0.05} + \frac{€2.00 \times \frac{10}{2} \times (0.20 - 0.05)}{0.12 - 0.05} \\
 &= €51.43.
 \end{aligned}$$

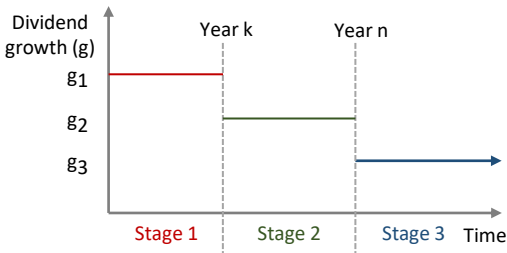
[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

4.

The value of common shares using the three-stage DDM

Recall: There are **two** popular versions of the three-stage DDM, distinguished by the modeling of the second stage.

4.1. The first version (the general three-stage model): The company is assumed to have three distinct stages of growth and the growth rate of the second stage is typically constant.



Three-stage DDM Valuation Version 1

$$V_0 = V_{S1} + V_{S2} + V_{S3}$$

$$= \sum_{t=1}^k \frac{D_0(1+g_1)^t}{(1+r)^t} + \sum_{t=k+1}^n \frac{D_0(1+g_1)^k(1+g_2)^{t-k}}{(1+r)^t} + \frac{D_0(1+g_1)^k(1+g_2)^{n-k}(1+g_3)}{(1+r)^n(r-g_3)}$$

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

4.

The value of common shares using the three-stage DDM

Example: Calculating value with the three-stage DDM

R&M has a current dividend of \$1.00 and a required rate of return of 12%. A dividend growth rate of 15% is projected for the next two years, followed by a 10% growth rate for the next four years before settling down to a constant 4% growth rate thereafter. Calculate the current value of R&M.

Answer:

We can use the formula of Three-stage DDM Valuation Version 1 from the previous slide to solve this example. However, in practice, we need to use a calculator, and as a first step, we have the following table:

Time	Value	Calculation	D_t or V_t
1	D_1	$\$1.00 \times 1.15$	\$1.150
2	D_2	$\$1.150 \times 1.15$	\$1.323
3	D_3	$\$1.323 \times 1.10$	\$1.455
4	D_4	$\$1.455 \times 1.10$	\$1.600
5	D_5	$\$1.600 \times 1.10$	\$1.760
6	D_6	$\$1.760 \times 1.10$	\$1.936
6	V_6	$\$1.936 \times 1.04 / (0.12 - 0.04)$	\$25.168

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

4.
The value of common shares using the three-stage DDM

Example: Calculating value with the three-stage DDM

Answer:

We can use the formula of Three-stage DDM Valuation Version 1 from the previous slide to solve this example. However, in practice, we need to use a calculator, and as a first step, we have the following table:

Now we enter the cash flows into our calculator, noting that the total cash flow at Time 6 is $\$1.936 + \$25.168 = \$27.104$:



CF0 = 0;

CO1 = 1.150;

CO2 = 1.323;

CO3 = 1.455;

CO4 = 1.600;

CO5 = 1.760;

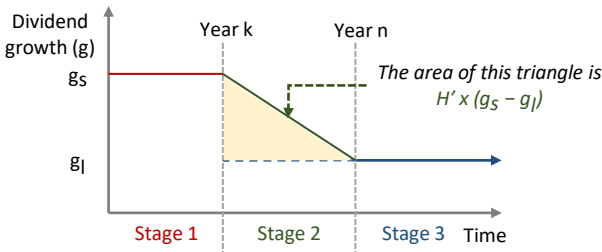
CO6 = 27.104; I = 12; CPT → NPV = \$18.864.

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

4.

The value of common shares using the three-stage DDM

4.2. The second version: The growth rate in the middle (second) stage is assumed to decline linearly to the mature growth rate: essentially, the second and third stages are treated as an H-model.



Three-stage DDM Valuation Version 2

$$\begin{aligned}
 V_0 &= V_{S1} + V_{S23} + V_{S2} \\
 &= \sum_{t=1}^k \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{D_0(1+g_s)^k(1+g_l)}{(1+r)^{k+1}} + \frac{D_0(1+g_s)^k \times H' \times (g_s - g_l)}{(1+r)^{k+1}(r - g_l)}
 \end{aligned}$$

where: $H' = \frac{n-k}{2}$ is half-life in years of the high-growth period (stage 2).

[LOS 2.n] Calculate and interpret the value of common shares using the two-stage DDM, the H-model, and the three-stage DDM

4.

The value of common shares using the three-stage DDM

Example: Three-stage growth model with linear growth decline in Stage 2

As an analyst, you have gathered the following information on a company you are tracking. The current annual dividend is \$0.75. Dividends are expected to grow at a rate of 12% over the next three years, decline linearly to 4% over the next six years, and then remain at a long-term equilibrium growth rate of 4% in perpetuity. The required return is 9%. Calculate the value of the company.

Answer:

Let's start by valuing the last two stages using the H-model. We know that:

$$\begin{aligned}
 V_3 &= \frac{D_0(1+g_S)^3(1+g_L)}{(r-g_L)} + \frac{D_0(1+g_S)^3 \times H' \times (g_S - g_L)}{(r-g_L)} \\
 &= \frac{\$0.75 \times (1+0.12)^3 \times (1+0.04)}{(0.09-0.04)} + \frac{\$0.75 \times (1+0.12)^3 \times \frac{6}{2} \times (0.12-0.04)}{(0.09-0.04)} \\
 &= \$26.9747.
 \end{aligned}$$

Now we have a series of three cash flows to discount in order to find the current value of the stock, and our financial calculator does the rest of the work:

$$\begin{aligned}
 CF_0 &= 0; CF_1 = 0.75 \times 1.12 = 0.84; CF_2 = 0.84 \times 1.12 = 0.9408; CF_3 = 0.9408 \times 1.12 + 26.9747 = 28.0284; I = 9; CPT \rightarrow NPV = \$23.2056.
 \end{aligned}$$

[LOS 2.o] Explain the use of spreadsheet modeling to forecast dividends and to value common shares

1.

Spreadsheet modeling

Problems with Two, Three-stage DDM and the H-model:

- A firm's dividends (or cash flows) often do not grow at a smooth rate for an extended period.
- When changes in dividends can be predicted, there can obviously be more than two or three stages of change involved.
- There are often idiosyncratic events that, even if they can be predicted, do not fit neatly into any of the patterns required by these models

In practice, a spreadsheet are much more likely to be used than any of the stylized models above when valuing equity securities due to its inherent flexibility and computational accuracy.

[LOS 2.o] Explain the use of spreadsheet modeling to forecast dividends and to value common shares

2.

The steps of using spreadsheet modeling

There are **four** steps of using spreadsheet modeling as the following:

01 Establish the base level of cash flows or dividends

In the case of dividends, this would ordinarily be either the amount paid over the preceding year or some normalized level based upon projected firm earnings.

02 Estimate changes in the firm's dividends for the foreseeable future and project future cash dividends on the basis of these estimates

03 Estimate what normalized level of growth will occur at the end of the supernormal growth period

This allows for an estimate of a terminal value, representing the cash flow (i.e., the firm's value if sold at this time) to be received at the end of the supernormal growth period.

04 Discount all projected dividends and the terminal value back to today to obtain an estimate of the firm's current value.

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

Based on any DDM, given current price and all inputs to a DDM except for the required return, an IRR can be calculated.

→ Such an IRR has been used as a required return estimate (an efficient market expected return).

1.1. Required return with the Gordon growth model:

If the dividend growth rate is constant forever, we can use the Gordon growth model to calculate the implied expected return given the expected dividend, the current market price, and the expected growth rate:

GGM's simplification

$$V_0 = \frac{D_0(1+g)}{r-g} \text{ or } V_0 = \frac{D_1}{r-g}$$

$$P_0 = V_0$$

Required return with GGM

$$r = \frac{D_1}{P_0} + g$$

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

Example: Calculating expected return with the Gordon growth model

Smyth & Weston Explosives' stock is expected to pay a dividend of \$1.60, has a current price of \$40.00, and has a projected growth rate of 9%. Calculate S&W's implied required return.

Answer:

Applying the formula of required return with GGM we have:

$$\begin{aligned} r &= \frac{D_1}{P_0} + g \\ &= \frac{\$1.60}{\$40.00} + 0.09 \\ &= 0.13 = 13\%. \end{aligned}$$

Thus, S&W's implied required return is 13%.

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

1.2. Required return with the H-model:

The H-model can be rewritten in terms of r and used to solve for r given the other model inputs:

H-model Valuation

$$V_0 = \frac{D_0(1 + g_L)}{r - g_L} + \frac{D_0 \times H \times (g_S - g_L)}{r - g_L}$$

$$P_0 = V_0$$

Required return with H-model

$$r = \frac{D_0}{P_0} \times [(1 + g_L) + H \times (g_S - g_L)] + g_L$$

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

Example: Solving for expected return with the H-model

Beluga Fisheries, Inc., just paid a current dividend of \$0.75, which has been growing at a rate of 10%. This growth rate is expected to decline to 5% over the next five years and then remain at 5% indefinitely. Calculate the implied required return for Beluga based on the current price of \$30.00.

Answer:

Applying the formula of required return with H-model we have:

$$\begin{aligned}
 r &= \frac{D_0}{P_0} \times (1 + g_I + H \times (g_S - g_I)) + g_I \\
 &= \frac{\$0.75}{\$30.00} \times (1 + 0.05 + \frac{5}{2} \times (0.10 - 0.05)) + 0.05 \\
 &= 0.0794 = 7.94\%.
 \end{aligned}$$

Thus, the implied required return for Beluga is 7.94%.

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

1.3. Required return with the general two-stage DDM:

Using the general two-stage model is more difficult because we have to solve for r with an iterative process because there is no closed-form solution.

Here is an example of how to approach the problem using the two-stage DDM.

Example: Solving for expected return with the two-stage DDM

Ozone Laboratories, Inc., recently paid a dividend of \$1.00. Dividends are expected to grow at a rate of 11% for the next two years and 8% thereafter. The implied required return for Ozone based on the current price of \$35.60 is closest to:

- A. 9.5%.
- B. 10.0%.
- C. 11.2%.

[LOS 2.i] Estimate a required return based on any DDM, including the Gordon growth model and the H-model

1.

Estimating a required return

Answer:

We have to solve for r in the following equation:

$$\frac{\$1.00 \times 1.11}{1+r} + \frac{\$1.00 \times 1.11^2}{(1+r)^2} + \frac{\$1.00 \times 1.11^2 \times 1.08}{(r-0.08)(1+r)^2} = \$35.60$$

This actually requires an iterative process. The easiest way to do this is to start with the middle value from the three answer choices (10%, in this case).

$$\text{Terminal value in Year 2 estimated: } P_2 = \frac{D_3}{r-g} = \frac{\$1.00 \times 1.11^2 \times 1.08}{0.10 - 0.08} = \$66.55$$

Use the calculator to value the stock:

$$CF_0 = 0; C_1 = 1.00 \times 1.11 = 1.11; C_2 = 1.00 \times 1.11^2 + 66.55 = 67.78;$$

$I = 10$; CPT $\rightarrow$ NPV = \$57.03 (too high relative to the given market price of \$35.60)

Hence, the discount rate has to be higher. The only higher value (from the multiple choices) is 11.2%. $\rightarrow$ C is correct.

$$\text{We can confirm this answer: } P_2 = \frac{D_3}{r-g} = \frac{\$1.00 \times 1.11^2 \times 1.08}{0.112 - 0.08} = \$41.59$$

$$CF_0 = 0; C_1 = 1.00 \times 1.11 = 1.11; C_2 = 1.00 \times 1.11^2 + 41.59 = 42.82;$$

$I = 11.2$; CPT $\rightarrow$ NPV = \$35.63 (close enough).

[LOS 2.p] Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis to estimate a company's sustainable growth rate

1.

Sustainable growth rate

The **sustainable growth rate (SGR)** is the rate at which earnings (and dividends) can continue to grow indefinitely, assuming that the firm's debt-to-equity ratio is unchanged and it doesn't issue new equity.

*SGR is a simple function of the **earnings retention ratio** and the **return on equity**:*

$$\text{SGR} = b \times \text{ROE}$$

where:

b is earnings retention rate = 1 – dividend payout rate

ROE is return on equity

The lower (higher) ROE, the lower (higher) SGR, all else constant.

The lower (higher) b, the lower (higher) SGR, all else constant.

→ This relationship has been called the *dividend displacement of earnings*.

The SGR is important because it tells us how quickly a firm can grow with internally generated funds.

[LOS 2.p] Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis...

1.

Sustainable growth rate

Example: Calculating SGR

Biotechnica, Inc., is growing earnings at an annual rate of 9%. It currently pays out dividends equal to 20% of earnings. Biotechnica's ROE is 15%. Calculate its SGR.

Answer:

Applying the SGR formula we have:

$$\text{SGR} = b \times \text{ROE}$$

$$= (1 - 20\%) \times 15\%$$

$$= 12\%.$$

Thus, its SGR is 12%.

[LOS 2.p] Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis...

2.

DuPont analysis

ROE can be estimated with the **DuPont formula**, which presents the relationship between **net profit margin**, **total assets turnover**, and **financial leverage** as determinants of ROE:

DuPont formula

$$ROE = \frac{NI}{E} = \frac{NI}{S} \times \frac{S}{A} \times \frac{A}{E}$$

where:

NI is net income

E is stockholder's equity

S is sales

A is total assets

[LOS 2.p] Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis...

2.

DuPont analysis

If the other factors remain constant, we can see that the growth of a firm's earnings (and dividends) is a function of its ROE and its retention rate:

SGR formula with DuPont analysis

$$g = \frac{NI - D}{NI} \times \frac{NI}{E} = \frac{NI - D}{NI} \times \frac{NI}{S} \times \frac{S}{A} \times \frac{A}{E}$$

The formula above has also been called the **PRAT** model, where SGR is a function of the firm's **financing decisions** (the retention rate (R), the financial leverage (T)) and the firm's **performance** (the profit margin (P), the the asset turnover (A)).

If the actual growth rate is forecasted to be greater than SGR, the firm will have to issue equity unless the firm increases its retention ratio, profit margin, total asset turnover, or leverage.

Note: We use beginning-of-year balance sheet values for mixed ratios (e.g., total asset turnover) unless otherwise specified in the question.

[LOS 2.p] Calculate and interpret the sustainable growth rate of a company and demonstrate the use of DuPont analysis...

2.

DuPont analysis

Example: Calculating ROE and SGR

Halo Construction has been successful in a mature industry. Over the last three years, Halo has averaged a profit margin of 10%, a total asset turnover of 1.8, and a leverage ratio of 1.25. Assuming Halo continues to distribute 40% of its earnings as dividends, calculate its long-term SGR.

Answer:

Applying the PRAT model we have:

$$g = P \times R \times A \times T$$

$$= 0.10 \times (1 - 0.4) \times 1.80 \times 1.25$$

$$= 0.135 = 13.5\%$$

Thus, its long-term SGR is 13.5%.

[LOS 2.j] Evaluate whether a stock is overvalued, fairly valued, or undervalued by the market based on a DDM estimate of value

1.

Evaluating value of a stock

We compare the price at which the stock is trading (Market price) and the price implied by a dividend discount model (Model price):

- Market price > Model price → Overvalued
- Market price < Model price → Undervalued
- Market price = Model price → Fairly valued

MODULE 3 : FREE CASH FLOW VALUATION

LEARNING OUTCOMES

- 3.a.** Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation
- 3.b.** Explain the ownership perspective implicit in the FCFE approach
- 3.c.** Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), earnings before interest, taxes, depreciation, and amortization (EBITDA), and cash flow from operations (CFO) to calculate FCFF and FCFE
- 3.d.** Calculate FCFF and FCFE
- 3.e.** Describe approaches for forecasting FCFF and FCFE
- 3.f.** Explain how dividends, share repurchases, share issues, and changes in leverage may affect future FCFF and FCFE
- 3.g.** Compare the FCFE model and dividend discount models
- 3.h.** Evaluate the use of net income and EBITDA as proxies for cash flow in valuation
- 3.i.** Explain the use of sensitivity analysis in FCFF and FCFE valuations

LEARNING OUTCOMES

- 3.j.** Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and justify the selection of the appropriate model given a company's characteristics
- 3.k.** Estimate a company's value using the appropriate free cash flow model(s)
- 3.l.** Describe approaches for calculating the terminal value in a multistage valuation model
- 3.m.** Evaluate whether a stock is overvalued, fairly valued, or undervalued based on a free cash flow valuation model

[LOS 3.a] Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation

1.

Warm-up: Free Cash Flow

Free cash flow (FCF) is the cash generated from a company's operations after deducting all of its capital expenditures, including investments in property, plant, and equipment, and is available for distribution to shareholders.

Free Cash Flow (FCF)

Free Cash Flow to the Firm (FCFF)

The cash available to all of the firm's investors, including stockholders and bondholders, after the firm buys and sells products, provides services, pays its cash operating expenses, and makes short- and long-term investments.

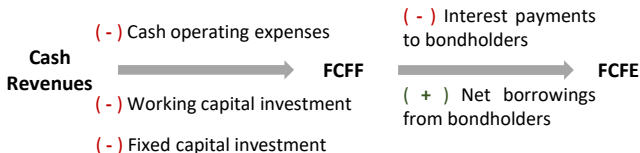
Free Cash Flow to Equity (FCFE)

The cash available to common shareholders after funding capital requirements, working capital needs, and debt financing requirements.

[LOS 3.a] Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation

1.

Warm-up: Free Cash Flow



 Why does net borrowing affect FCFE but not FCFF?

- Think of **FCFF** as the cash flow generated by the firm's **core business**. Borrowing isn't generated by the firm's core business, so net borrowing has no impact on FCFF.
- Think of **FCFE** as cash that could be **given to shareholders**, if management wanted to, so net borrowing increases FCFE.

[LOS 3.a] Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation

2.

Compare FCFF and FCFE

*We've learnt the definitions of FCFF and FCFE. However, another key aspect is understanding **which cash flows** to discount and at **which rate** to estimate which value.*

Free Cash Flow to the Firm (FCFF)

The value of the firm is the present value of the expected future **FCFF** discounted at the **WACC**:

Firm value = FCFF discounted at the WACC.

→ **Equity value = Firm value – market value of debt**

Free Cash Flow to Equity (FCFE)

The value of the firm's equity is the present value of the expected future **FCFE** discounted at the **required return on equity**:

Equity value = FCFE discounted at the required return on equity.

[LOS 3.a] Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation

2.

Compare FCFF and FCFE

 *In what cases should we use FCFF or FCFE to calculate equity value?*

- If the company's capital structure is relatively stable, using FCFE to value equity is more direct and simpler than using FCFF.
- If a company has negative FCFE and volatile capital structure, FCFF is generally the best choice.

When a firm's capital structure is **volatile**, the **adjusted present value (APV)** approach can also be used to facilitate the equity valuation.

Adjusted present value (APV)

Firm value

The value of the company under the assumption that debt is not used (i.e., unlevered firm value)



The NPV of any effects of debt on firm value (such as any tax benefits of using debt and any costs of financial distress)

[LOS 3.a] Compare the free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) approaches to valuation

2.

Compare FCFF and FCFE

	FCFF	FCFE
Definition	Cash flow generated by the firm's operations that is in excess of the capital investment required to sustain the firm's current productive capacity	Cash available to stockholders after funding capital requirements and expenses associated with debt financing
Discount rate	WACC	Required return on equity (r)
Equity value	FCFF discounted at the WACC - market value of debt	FCFE discounted at r
Used when	Negative FCFE and/or volatile capital structure	Capital structure is relatively stable

[LOS 3.b] Explain the ownership perspective implicit in the FCFE approach

1.

Ownership perspective implicit in the FCFE approach

The ownership perspective in the free cash flow approach is that of:

- An acquirer who can change the firm's dividend policy, which is a control perspective, or
- Minority shareholders of a company that is in-play (i.e., it is a takeover target with potential bidders).

*The free cash flow valuation is often preferred over dividend-based valuation for **four** reasons as the following:*

01

Many firms pay no, or low, cash dividends.

02

Dividends are paid at the discretion of the board of directors. It may, consequently, be poorly aligned with the firm's long-run profitability.

03

If a company is viewed as an acquisition target, free cash flow is a more appropriate measure because the new owners will have discretion over its distribution (control perspective).

04

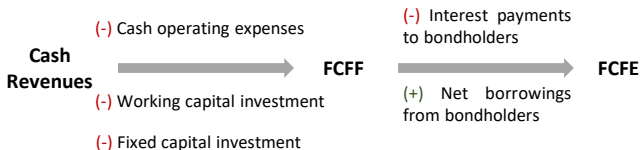
Free cash flows may be more related to long-run profitability of the firm as compared to dividends.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), earnings before interest, taxes, depreciation, and amortization (EBITDA), and cash flow from operations (CFO) to calculate FCFF and FCFE

1.

Relationship between FCFF and FCFE

We have learnt about this following figure, which illustrates the relationship between Cash revenues, FCFF and FCFE:



If we start with FCFF, we have to adjust for the two cash flows to bondholders to calculate FCFE:

- The after-tax interest expense
- Any new long- or short-term borrowings.

We only subtract the after-tax interest expense because paying interest reduces the firm's tax bill and reduces the cash available to the shareholders by the interest paid minus the taxes saved.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

1.

Relationship between FCFF and FCFE

Calculating FCFE from FCFF

$$\text{FCFE} = \text{FCFF} - \text{Int} \times (1 - \text{tax rate}) + \text{Net borrowing}$$

where:

Int = Interest expense

Net borrowing = Long- and short-term new debt issues – long- and short-term debt repayments

Given the relationship between FCFF and FCFE, we now focus on calculating FCFF from net income, earnings before interest and taxes (EBIT), earnings before interest, taxes, depreciation, and amortization (EBITDA), and cash flow from operations (CFO) because it's relatively straightforward to derive FCFE from these figures.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

 **Recall:** FCFF is the cash flow available to the company's suppliers of capital (stockholders and bondholders) after all operating expenses (including taxes) have been paid and operating investments have been made.

*Keeping in mind that a **noncash charge** is a charge or expense that **does not involve the outlay of cash**, we can write the expression for FCFF as follows:*

FCFF = Net income available to common shareholders (NI)

Plus: Noncash charges (NCC) **(1)**

Plus: Interest expense $\times (1 - \text{Tax rate})$ **(2)**

Less: Investment in fixed capital (FCInv) **(3)**

Less: Investment in working capital (WCInv) **(4)**

Calculating FCFF from NI

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

*We will consider these **four** figures in detail in the next slides.*

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.1. Noncash charges

Noncash charges are added back to net income to arrive at FCFF because they represent expenses that reduced reported net income but didn't actually result in an outflow of cash.

Here are some examples of noncash charges how to adjust them from net income:

Noncash Item	Adjustment to NI to arrive at FCFF
Depreciation & Amortization expense	Added back
Restructuring charges (expense/income)	Added back/Subtracted
Amortization of long-term bond discounts/premiums	Added back/Subtracted
Losses/Gains on non-operating activity	Added back/Subtracted
Deferred taxes	Added back but calls for special attention

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.2. Interest expense

Interest was expensed on the income statement, but it represents a financing cash flow to bondholders that is available to the firm before it makes any payments to its capital suppliers.

→ We have to add it back to net income to estimate FCFF.

We don't add back the entire interest expense, only the after-tax interest cost, because paying interest reduces our tax bill.

Example: if the marginal tax rate is 30%, every dollar of interest paid reduces the tax bill by 30 cents. The net effect on free cash flow is an increase in the after-tax interest cost of 70 cents.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.3. Fixed capital investment

Investments in fixed capital do not appear on the income statement, but they do represent cash leaving the firm.

→ We have to subtract them from net income to estimate FCFF.



How to calculate Fixed capital investment (FCInv)

Fixed capital investment is a net amount: the difference between capital expenditures (investments in long-term fixed assets) and the proceeds from the sale of long-term assets.

- **If no long-term assets were sold during the year:**

FCInv is capital expenditures and will also equal the change in the gross PP&E account from the balance sheet.

Calculating FCInv

$$\begin{aligned} \text{FCInv} &= \text{ending gross PP\&E} - \text{beginning gross PP\&E} \\ &= \text{ending net PP\&E} - \text{beginning net PP\&E} + \text{depreciation} \end{aligned}$$

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.3. Fixed capital investment

- If long-term assets were sold during the year, we have:

Calculating FCInv

$$\text{FCInv} = \text{capital expenditures} - \text{proceeds from sales of long-term assets}$$

where:

- Determine **capital expenditures** from either (1) an item in the statement of cash flows called something like “purchase of fixed assets” or “purchases of PP&E” under cash flow from investing activities, or (2) data provided in the vignette.
- Determine **proceeds from sales of fixed assets** from either (1) an item in the statement of cash flows called something like “proceeds from disposal of fixed assets,” or (2) data provided in the vignette.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.3. Fixed capital investment

- **If long-term assets were sold during the year, we have:**

If capital expenditures or sales proceeds are not given directly, find gain (loss) on asset sales from the income statement and PP&E figures from balance sheet.

Calculating FCInv

$\text{FCInv} = \text{capital expenditures} - \text{proceeds from sales of long-term assets}$

Calculating FCInv

$\text{FCInv} = \text{ending net PP\&E} - \text{beginning net PP\&E} + \text{depreciation} + / - \text{loss/gain on sale}$

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

Example: Calculating FCInv with no long-term asset sales

Airbrush, Inc., financial statements for 2023 include the following information:

Selected Financial Data

	2023	2022
Gross PP&E	\$5,000	\$4,150
Accumulated depreciation	(\$1,500)	(\$1,200)
Net PP&E	\$3,500	\$2,950

There were no sales of PP&E during the year; depreciation expense was \$300. Calculate Airbrush's FCInv for 2023.

Answer:

We have:

FCInv = capital expenditures = ending gross PP&E – beginning gross PP&E = \$5,000 - \$4,150 = \$850.

Alternatively, FCInv = capital expenditures = ending net PP&E – beginning net PP&E + depreciation = \$3,500 - \$2,950 + \$300 = \$850.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

2.

Computing FCFF from Net Income

2.4. Working capital investment:

- The investment in net working capital is equal to the change in working capital.
- For cash flow and valuation purposes, working capital is defined as current assets minus current liabilities, excluding cash and short-term debt (which includes notes payable and the current portion of long-term debt).

Calculating working capital

Working capital = Current assets (excluding cash) – Current liabilities (excluding short-term debt)

Calculating WCInv

WCInv = ending working capital – beginning working capital

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

3.

Computing FCFF from EBIT and EBITDA

Recall: We learned about Net income (NI), Earnings before interest and taxes (EBIT), and Earnings before interest, taxes, depreciation, and amortization (EBITDA) as well as their relationship in CFA Level I.

Original formula

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

Calculating NI from EBIT

$$\text{NI} = (\text{EBIT} - \text{Int}) \times \text{tax rate} = \text{EBIT}(1 - \text{tax rate}) - \text{Int} \times (1 - \text{tax rate})$$

Calculating NI from EBITDA

$$\begin{aligned} \text{NI} &= (\text{EBITDA} - \text{Dep} - \text{Int}) (1 - \text{tax rate}) \\ &= \text{EBITDA} (1 - \text{tax rate}) - (\text{Int} + \text{Dep}) \times (1 - \text{tax rate}) \end{aligned}$$

Replacing the new NI into the original formula

Calculating FCFF from EBIT & EBITDA

$$\begin{aligned} \text{FCFF} &= \text{EBIT} \times (1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv} \\ \text{FCFF} &= \text{EBITDA} \times (1 - \text{tax rate}) + \text{Dep} \times \text{tax rate} - \text{FCInv} - \text{WCInv} \end{aligned}$$

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

4.

Computing FCFF from CFO



Recall: We learned about Net income (NI), Cash flow from operations (CFO), as well as their relationship in CFA Level I.

Calculating CFO from NI

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WCInv} \rightarrow \text{NI} = \text{CFO} - \text{NCC} + \text{WCInv}$$

Calculating FCFF from CFO

$$\begin{aligned} \text{FCFF} &= (\text{NI} + \text{NCC} - \text{WCInv}) + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} \\ &= \text{CFO} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} \end{aligned}$$

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

5.

Computing FCFE from NI and CFO



Recall: The relationship between FCFF and FCFE is:

$$\text{FCFE} = \text{FCFF} - \text{Int} \times (1 - \text{tax rate}) + \text{Net borrowing}$$

Thus, from the formulas of FCFE with NI and CFO, we can easily derive the formulas of FCFF with these figures as the following:

Calculating FCFE from NI

$$\text{FCFE} = \text{NI} + \text{NCC} + \text{Net borrowing} - \text{FCInv} - \text{WCInv}$$

Calculating FCFE from CFO

$$\text{FCFE} = \text{CFO} + \text{Net borrowing} - \text{FCInv}$$

Tip: We can easily derive FCFE from NI and CFO by replacing “Int x (1 – tax rate)” with “Net borrowing” from the FCFF formulas.

[LOS 3.c] Explain the appropriate adjustments to net income, earnings before interest and taxes (EBIT), ...

6.

Free Cash Flow With Preferred Stock

The FCFF and FCFE formulas previously assume that the company uses only debt and common equity to raise funds.

We need revise the FCFF and FCFE formulas to reflect the payment of preferred dividends and any issuance or repurchase of such shares.

Note: Treat preferred stock just like debt, except preferred dividends are not tax deductible.

 *How should FCFF, FCFE, and WACC change after adjusting for preferred stock*

FCFF

Any preferred dividends should be added back to the FCFF in the net income approach to generating FCFF.

FCFE

Net borrowing should be modified to reflect new debt borrowing and net issuances by the amount of the preferred stock.

WACC

WACC should be revised to reflect the percent of total capital raised by preferred stock and the cost of that capital source.

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

We will illustrate some examples to see if all these formulas really work

Example 1: Calculating FCFF from NI and FCFE from FCFF

Anson Ford, CFA, is analyzing the financial statements of Sting's Delicatessen. He has a 20X6 income statement and balance sheet, as well as 20X7 income statement, balance sheet, and cash flow from operations forecasts (as shown in the following tables). Assume there will be no sales of long-term assets in 20X7.

Calculate forecasted free cash flow to the firm (FCFF) and free cash flow to equity (FCFE) for 20X7.

Note: See *Sting's Income Statement*, *Sting's Balance Sheet*, *Sting's Cash Flow From Operations Forecast* in the next slides

[LOS 3.d] Calculate FCFF and FCFE

1. Calculating FCFF and FCFE

Sting's Income Statement

	20X7 Forecast	20X6 Actual
Sales	\$300	\$250
Cost of goods sold	(120)	(100)
Gross profit	180	150
SG&A	(35)	(30)
Depreciation	(50)	(40)
EBIT	95	80
Interest expense	(15)	(10)
Pre-tax earnings	80	70
Taxes (at 30%)	(24)	(21)
Net income	\$56	\$49

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Sting's Balance Sheet

	20X7 Forecast	20X6 Actual
Cash	\$10	\$5
Accounts receivable	30	15
Inventory	40	30
Current assets	\$80	\$50
Gross PP&E	400	300
Accumulated depreciation	(190)	(140)
Total Assets	\$290	\$210

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Sting's Balance Sheet - Assets

	20X7 Forecast	20X6 Actual
Cash	\$10	\$5
Accounts receivable	30	15
Inventory	40	30
Current assets	\$80	\$50
Gross PP&E	400	300
Accumulated depreciation	(190)	(140)
Total Assets	\$290	\$210

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Sting's Balance Sheet – Liabilities and owners' equity

	20X7 Forecast	20X6 Actual
Accounts payable	\$20	\$20
Short-term debt	20	10
Current liabilities	\$40	\$30
Long-term debt	114	100
Common stock	50	50
Retained earnings	86	30
Total liabilities and owners' equity	\$290	\$210

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Sting's Cash Flow From Operations Forecast

	20X7 Forecast
Net income	\$56
+ depreciation	50
- WCInv	25
Cash flow from operations	\$81

Answer:

According to the formula of FCFF calculation from NI we have:

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

From Cash Flow From Operations Forecast table, we can determine that NI = 56 and depreciation is the only NCC = 50.

From Income Statement table, we can determine that Int = 15 and tax rate = 30%.

Thus, we need to calculate FCInv and WCInv to derive FCFF.

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Answer:

According to the formulas of FCInv and WCInv we have:

$$\text{FCInv} = \text{ending gross PP\&E} - \text{beginning gross PP\&E}$$

$$= 400 - 300 \text{ (see Balance Sheet table)}$$

$$= 100$$

$$\begin{aligned} \text{WCInv} &= \text{ending working capital} - \text{beginning working capital} = (\text{ending} \\ &\text{Accounts receivable} + \text{ending Inventory} - \text{ending Accounts payable}) - \\ &(\text{beginning Accounts receivable} + \text{beginning Inventory} - \text{beginning Accounts} \\ &\text{payable}) \end{aligned}$$

$$= (30 + 40 - 20) - (15 + 30 - 20) \text{ (see Balance Sheet table)}$$

$$= 25$$

Now, we have:

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv} - \text{WCInv}$$

$$= 56 + 50 + 15 \times (1 - 0.3) - 100 - 25$$

$$= -\$8.5.$$

Note: It's entirely possible that FCFF can be negative in the short term. We'll talk more later about how to value firms with negative FCFF.

[LOS 3.d] Calculate FCFF and FCFE

1.

Calculating FCFF and FCFE

Answer:

Next, according to the formula of FCFE calculation from FCFF we have:

$$FCFE = FCFF - \text{Int} \times (1 - \text{tax rate}) + \text{Net borrowing}$$

Recall the formula of Net borrowing that Net borrowing is the difference between the new debt issues and debt repayments, and is calculated as the change in long- and short-term debt:

Net borrowing = (ending long- and short-term debt) – (beginning long- and short-term debt)

$$= (114 + 20) - (100 + 10) \text{ (see Balance Sheet table)}$$

$$= 24$$

Thus, $FCFE = FCFF - \text{Int} \times (1 - \text{tax rate}) + \text{Net borrowing}$

$$= -8.5 - 15 \times (1 - 0.3) + 24$$

$$= \$5.$$

[LOS 3.d] Calculate FCFF and FCFE

1. Calculating FCFF and FCFE

Example 2: Calculating FCFF and FCFE with the other formulas

Using the data of Example 1, calculate FCFF starting with EBIT, EBITDA, and CFO, and calculate FCFE starting with NI and CFO.

Answer:

According to the other formulas that we have learnt, we have:

$$\text{FCFF} = \text{EBIT} \times (1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$$

$$= 95 \times (1 - 0.3) + 50 - 100 - 25 = -\$8.5$$

$$\text{FCFF} = \text{EBITDA} \times (1 - \text{tax rate}) + \text{Dep} \times \text{tax rate} - \text{FCInv} - \text{WCInv}$$

$$= 145 \times (1 - 0.3) + 50 \times 0.3 - 100 - 25 = -\$8.5$$

$$\text{FCFF} = \text{CFO} + \text{Int} \times (1 - \text{tax rate}) - \text{FCInv}$$

$$= 81 + 15 \times (1 - 0.3) - 100 = -\$8.5$$

$$\text{FCFE} = \text{NI} + \text{NCC} + \text{Net borrowing} - \text{FCInv} - \text{WCInv}$$

$$= 56 + 50 + 24 - 100 - 25 = \$5$$

$$\text{FCFE} = \text{CFO} + \text{Net borrowing} - \text{FCInv}$$

$$= 81 - 100 + 24 = \$5.$$

[LOS 3.d] Calculate FCFF and FCFE

2. The uses of free cash flow

- An analyst may also be concerned about the uses of cash flow.
- Typically, this is done to **verify the FCFE and FCFF** calculation, as FCFF sources must always equal FCFF uses, and FCFE sources must always equal FCFE uses.

2.1. The uses of FCFF

We calculate uses of FCFF as follows:

Uses of FCFF = Increases (or minus decreases) in cash balances

Plus: **Net payments to providers of debt capital**, which are calculated as:

- Plus: Interest expense $\times (1 - \text{Tax rate})$.
- Plus: Repayment of principal in excess of new borrowing (or minus new borrowing in excess of debt repayment if new borrowing is greater).

Plus: **Payments to providers of equity capital**, which are calculated as:

- Plus: Cash dividends.
- Plus: Share repurchases in excess of share issuance (or minus new share issuance in excess of share repurchases if share issuance is greater).

Note: Uses of FCFF must equal sources of FCFF as previously calculated.

[LOS 3.d] Calculate FCFF and FCFE

2. The uses of free cash flow

2.2. The uses of FCFE

We calculate uses of FCFE as follows:

Uses of FCFE = Increases (or minus decreases) in cash balances

Plus: **Payments to providers of equity capital**, which are calculated as:

- Plus: Cash dividends.
- Plus: Share repurchases in excess of share issuance (or minus new share issuance in excess of share repurchases if share issuance is greater).

Note: Uses of FCFE must equal sources of FCFE as previously calculated.

[LOS 3.e] Describe approaches for forecasting FCFF and FCFE

1. Approaches for forecasting FCFF and FCFE

There are **two** common approaches used to forecast future FCFF and FCFE.

1.1. The first method: Applying some constant growth rate to a current level of free cash flow

This approach is appropriate if a company's free cash flow has tended to grow at a constant rate and if historical relationships between free cash flow and fundamental factors are expected to continue.

→ This is the same method we used for dividend discount models.

Example: Use Pitts Corporation data to compute its FCFF for the next three years. Assume growth in FCFF remains at the historical levels of 15% a year. The answer is as follows (in millions):

	2020 Actual	2021 Estimate	2022 Estimate	2023 Estimate
FCFF	155.00	178.25	204.99	235.74

Note: The growth rate for FCFF is usually different than the growth rate for FCFE.

[LOS 3.e] Describe approaches for forecasting FCFF and FCFE

1. Approaches for forecasting FCFF and FCFE

1.2. The second method: Forecasting the underlying components of free cash flow and calculate each year separately

This approach is a more realistic, more flexible, and more complicated method because we can assume that each component of free cash flow is growing at a different rate over some short-term horizon.

*In this discussion, we illustrate a simple **sales-based forecasting method** for FCFF and FCFE based on the following major assumption:*

- Investment in fixed capital in excess of depreciation (FCInv – Dep) and investment in working capital (WCInv) both bear a constant relationship to forecast increases in the size of the company as measured by increases in sales.
- The firm maintains a target debt-to-asset ratio (DR) for net new investment in fixed capital and working capital.

1.2.1. Forecasting FCFF:

 **Recall:** Calculating FCFF from EBIT:

$$\text{FCFF} = \text{EBIT} \times (1 - \text{tax rate}) + \text{Dep} - \text{FCInv} - \text{WCInv}$$

[LOS 3.e] Describe approaches for forecasting FCFF and FCFE

1. Approaches for forecasting FCFF and FCFE

In the case of FCFF forecasting, we need to estimate FCInv and WCInv by multiplying their past proportion to sales increases by the forecasted sales increases.

Incremental fixed capital expenditures as a proportion of sales increases are computed as follows:

$$\alpha = \frac{\text{Capital expenditures} - \text{Depreciation expense}}{\text{Increase in sales}}$$

Similarly, incremental working capital expenditures as a proportion of sales increases are:

$$\beta = \frac{\text{Increase in working capital}}{\text{Increase in sales}}$$

- Estimate FCInv: $(\text{Capital expenditures} - \text{Depreciation expense})_t = \alpha_{t-1} \times \text{Increase in sales}_t$
- Estimate WCInv: $\text{Increase in working capital}_t = \beta_{t-1} \times \text{Increase in sales}_t$

[LOS 3.e] Describe approaches for forecasting FCFF and FCFE

1. Approaches for forecasting FCFF and FCFE

1.2.1. Forecasting FCFE:

 **Recall:** Calculating FCFE from NI:

$$\text{FCFE} = \text{NI} - (\text{FCInv} - \text{Dep}) - \text{WCInv} + \text{Net borrowing}$$

In the case of FCFE forecasting, we need to estimate FCInv and WCInv as well, and make an assumption that a specified percentage of the sum of FCInv – Dep and WCInv is financed based on a target DR.

Calculating Net borrowing

$$\text{Net borrowing} = \text{DR} \times (\text{FCInv} - \text{Dep}) + \text{DR} \times \text{WCInv}$$

Calculating Net borrowing

$$\text{FCFE} = \text{NI} - (1 - \text{DR}) \times (\text{FCInv} - \text{Dep}) - (1 - \text{DR}) \times \text{WCInv}$$

where:

DR is target debt-to-asset ratio.

[LOS 3.f] Explain how dividends, share repurchases, share issues, and changes in leverage may affect future FCFF and FCFE

1.

Effect of Financing Decisions on Free Cash Flow

Dividends, share repurchases, and share issues represent *uses* of FCFE and FCFF.

These financing decisions don't affect the level of cash flow available.



What about changes in leverage?

- **If leverage increases:**

Effect on FCFE: Net borrowing increases in the current year and interest expense increases in futures years → FCFE increases in the current year, but decreases in future years.

Effect on FCFF: Because EBIT is unchanged → No effect on FCFF.

- **If leverage decreases:**

Similarly, FCFE decreases in the current year, but increases in future years, and there is no effect on FCFF.

[LOS 3.f] Explain how dividends, share repurchases, share issues, and changes in leverage may affect future FCFF and FCFE

1.

Effect of Financing Decisions on Free Cash Flow

	FCFF	FCFE
Dividends	None	None
Share repurchase	None	None
Share issue	None	None
Change in leverage	None	Short-term and long-term effects partially offset

[LOS 3.g] Compare the FCFE model and dividend discount models

1. Comparing the FCFE model and dividend discount models

If all the inputs were known and mutually consistent (one possibility would be that **FCFE equals cash dividends** each year), a DDM and an FCFE model would result in **identical valuations** for a stock.

Generally, FCFE and cash dividends are not equal, but the same economic forces that lead to **low (high) dividends** lead to **low (high) FCFE**.

*FCFE and dividends will usually **differ** as follows:*

	FCFE model	DDM
Perspective	Control perspective	Minority perspective
Value recognition	Immediate	Only when dividend policy accurately reflects the firm's long-run profitability

[LOS 3.h] Evaluate the use of net income and EBITDA as proxies for cash flow in valuation

1.

Use of Net income as proxies for FCFE in valuation

 **Recall:** We learned the calculation of FCFE from NI:

$$\text{FCFE} = \text{NI} + \text{NCC} - \text{FCInv} - \text{WCInv} + \text{net borrowing}$$

We can see that:

- Net income includes noncash charges like depreciation that have to be added back to arrive at FCFE.
- Net income ignores cash flows that don't appear on the income statement, such as investments in working capital and fixed assets as well as net borrowings.

Net income is a poor proxy for FCFE.

[LOS 3.h] Evaluate the use of net income and EBITDA as proxies for cash flow in valuation

2.

Use of EBITDA as proxies for FCFF in valuation

 **Recall:** We learned the calculation of FCFF from EBITDA:

$$\text{FCFF} = \text{EBITDA} \times (1 - \text{tax rate}) + \text{Dep} \times \text{tax rate} - \text{FCInv} - \text{WCInv}$$

We can see that:

- EBITDA doesn't reflect the cash taxes paid by the firm.
- EBITDA ignores the cash flow effects of the investments in working capital and fixed capital.

EBITDA is a poor proxy for FCFF.

[LOS 3.i] Explain the use of sensitivity analysis in FCFF and FCFE valuations

1.

Use of sensitivity analysis in FCFF and FCFE valuations

Sensitivity analysis shows how sensitive the valuation results are to changes in each of a model's inputs.

There are **two** major sources of error in valuation analysis:

01

Estimating the future growth in FCFF and FCFE.

Growth forecasts depend on a firm's future profitability, which in turn depends on sales growth, changes in profit margin, position in the life cycle, its competitive strategy, and the overall profitability of the industry.

02

The chosen base years for the FCFF or FCFE growth forecasts.

Given the same required rates of return and growth rates, the value of the firm or the value of equity will increase or decrease proportionately with the initial value of FCFF or FCFE used.

[LOS 3.i] Explain the use of sensitivity analysis in FCFF and FCFE valuations

1.

Use of sensitivity analysis in FCFF and FCFE valuations

Example: Sensitivity Analysis of an FCFE Valuation

Antonio Sousa is valuing the equity of Petroleo Brasileiro, commonly known as Petrobras, by using the single-stage (constant-growth) FCFE model. Estimated FCFE per share for the year just ended is 2.59 Brazilian reais (BRL). Sousa's best estimates of input values for the analysis are as follows:

- The FCFE growth rate is 7.0%.
- The risk-free rate is 8.9%.
- The equity risk premium is 5.3%.
- Beta is 1.4.

Analysis:

Using the capital asset pricing model (CAPM), Sousa estimates that the required rate of return for Petrobras is:

$$r = \text{risk-free rate} + \text{beta} \times \text{equity risk premium} = 8.9\% + 1.4 \times 5.3\% = 16.32\%.$$

Thus, the estimated value per share is:

$$V_0 = \frac{\text{FCFE}_0 \times (1 + g)}{r - g} \text{ (similar to GGM, we will discuss this formula in detail in the next Los)}$$

$$= \frac{2.59 \times (1 + 0.07)}{0.1632 - 0.07} = \text{BRL}29.73.$$

[LOS 3.i] Explain the use of sensitivity analysis in FCFF and FCFE valuations

1.

Use of sensitivity analysis in FCFF and FCFE valuations

Example: Sensitivity Analysis of an FCFE Valuation

Analysis:

The table below shows Sousa's base case and the highest and lowest reasonable alternative estimates.

The column "Valuation with Low Estimate" gives the estimated value of Petrobras based on the low estimate for the variable on the same row of the first column and the base-case estimates for the remaining three variables. "Valuation with High Estimate" gives a similar estimated value based on the high estimate for the variable at issue.

Variable	Base case estimate	Low estimate	High estimate	Low Valuation	High Valuation
Beta	1.4	1.2	1.6	BRL33.55	BRL26.70
Risk-free rate	8.9%	7.9%	9.9%	BRL33.31	BRL26.85
Equity risk premium	5.3%	4.3%	6.3%	BRL34.99	BRL25.85
FCFE growth rate	7.0%	5.0%	9.0%	BRL24.02	BRL38.57

[LOS 3.i] Explain the use of sensitivity analysis in FCFF and FCFE valuations

1.

Use of sensitivity analysis in FCFF and FCFE valuations

Analysis:

Variable	Base case estimate	Low estimate	High estimate	Low Valuation	High Valuation
Beta	1.4	1.2	1.6	BRL33.55	BRL26.70
Risk-free rate	8.9%	7.9%	9.9%	BRL33.31	BRL26.85
Equity risk premium	5.3%	4.3%	6.3%	BRL34.99	BRL25.85
FCFE growth rate	7.0%	5.0%	9.0%	BRL24.02	BRL38.57

As table above shows, the value of Petrobras is very sensitive to the inputs. The value is **negatively** related to changes in the **beta**, the **risk-free rate**, and the **equity risk premium** and **positively** related to changes in the **FCFE growth rate**.

Of the four variables considered, the stock valuation is most sensitive to the range of estimates for the FCFE growth rate (a range from BRL24.02 to BRL38.57).

[LOS 3.j] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and justify the selection of the appropriate model given a company's characteristics

1.

Single-stage FCFF model

The **single-stage FCFF model** is analogous to the Gordon growth model discussed in the Module 2 on dividend valuation models. It is useful for stable firms in mature industries.

Model assumptions

FCFF grows at a constant rate (g) forever.

(g) is less than the weighted average cost of capital (WACC).

 **Recall:** We learned about WACC in CFA Level I: the weighted average of the rates of return required by each of the capital suppliers (usually just equity and debt) .

WACC will change over time as the firm's capital structure changes. Therefore, target capital structure weights are usually used rather than actual weights.

[LOS 3.j] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

1.

Single-stage FCFF model

$$WACC = w_e \times r_e + w_d \times r_d \times (1 - \text{tax rate})$$

where:

$$w_e = \frac{\text{market value of equity}}{\text{market value of equity} + \text{market value of debt}}$$

$$w_d = \frac{\text{market value of debt}}{\text{market value of equity} + \text{market value of debt}}$$

Value of the firm

$$V_f = \frac{FCFF_1}{WACC - g} = \frac{FCFF_0(1 + g)}{WACC - g}$$

where:

$FCFF_1$ is the expected free cash flow to the firm in one year

$FCFF_0$ is the starting level of FCFF

g is the constant expected growth rate in FCFF

WACC is the weighted average cost of capital

[LOS 3.j] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

2.

Single-stage FCFE model

The **single-stage constant-growth FCFE valuation model** is analogous to the singlestage FCFF model, with FCFE instead of FCFF and required return on equity instead of WACC.

Value of equity

$$V_e = \frac{FCFE_1}{r - g} = \frac{FCFE_0(1 + g)}{r - g}$$

where:

$FCFE_1$ is the expected free cash flow to equity in one year

$FCFE_0$ is the starting level of FCFE

g is the constant expected growth rate in FCFE

r is the required return on equity

Note: It's quite likely that a firm's growth rate in FCFF will be different than its FCFE growth rate.

The single-stage FCFE model is often used in international valuation, especially for companies in countries with **high inflationary expectations**.

→ **Real values** are estimated for the inputs to the single-stage FCFE model.

[LOS 3.] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

3.

Multistage Models: How many variations are there?

 **Recall:** We learned about Multistage models of dividend discounting in Module 2 such as Two-stage, H-model, Three-stage.

In general, they can be applied to free cash flow valuation, except H-model, because it is essentially a special case of the Two-stage model.

*There are at least **three** important ways that these models can differ:*

01

FCFF versus FCFE.

- The value of the firm: using FCFF discounted at the WACC.
- The value of equity: using FCFE discounted at r

02

Two-stage versus three-stage models.

It depends on how we model the future growth pattern.

03

Forecasting growth in total free cash flow (FCFF or FCFE) versus forecasting the growth rates in the components of free cash flow.

Unlike multistage DDMs, in which the growth rates are consistently dividend growth rates, in free cash flow models, the “growth rate” may refer to different variables: FCF, NI, Sales,...

[LOS 3.] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

4.

Two-Stage FCFF and FCFE Models

Let's discuss some examples of two-stage models. For now, concentrate on the differences in the assumptions: FCFF versus FCFE, growth pattern in the first stage, and forecasting total free cash flow versus forecasting its components.

We can analyze a:

- Two-stage FCFF model in which FCFF is projected to grow at 20% for the first four years and then 4% every year thereafter.
- Two-stage FCFE model in which FCFE declines from 20% to 4% over four years and then stays at 4% forever.
- Two-stage FCFE model in which sales grow at 20% for four years, the net profit margin is constant at 8%, fixed capital investment is equal to 60% of the dollar increase in sales, working capital investment is equal to 25% of the dollar increase in sales, and the debt ratio is 50%. Given a starting value for sales, we have all we need to forecast FCFE for the first four years.

Note: We also need a **terminal value** at the end of the first growth stage for each of these examples by applying a single-stage free cash flow model at the point in time when growth settles down to its long-run level.

[LOS 3.j] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

4.

Two-Stage FCFF and FCFE Models

A general expression for the two-stage FCFF valuation model is:

Firm valuation

$$V_f = \sum_{t=1}^n \frac{FCFF_t}{(1 + WACC)^t} + \frac{FCFF_{n+1}}{(WACC - g)(1 + WACC)^n}$$

The summation gives the present value of the first n years of FCFF. The terminal value of the FCFF from Year $n + 1$ forward is $\frac{FCFF_{n+1}}{WACC - g}$, which is discounted at the WACC for n periods to obtain its present value.

Similarly, a general expression for the two-stage FCFE valuation model is:

Equity valuation

$$V_e = \sum_{t=1}^n \frac{FCFE_t}{(1 + r)^t} + \frac{FCFE_{n+1}}{(r - g)(1 + r)^n}$$

where:

$\frac{FCFE_{n+1}}{r - g}$ is the terminal value of the FCFE from Year $n + 1$ forward in Year n .

[LOS 3.j] Explain the single-stage (stable-growth), two-stage, and three-stage FCFF and FCFE models and...

5.

Three-Stage FCFF and FCFE Models

Three-stage models have all the complications of the two-stage models, with an additional growth stage to consider.

We can analyze a:

- Three-stage FCFE model in which FCFE grows at 30% for two years (Stage 1), 15% for four years (Stage 2), and then 5% forever (Stage 3).
- Three-stage FCFF model in which FCFF grows at 25% for three years (Stage 1), declines to 4% over next the five years (Stage 2), then stays at 4% forever (Stage 3).
- Three-stage FCFE model in which we forecast the components of FCFE over three different stages.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

1.

Single-stage FCFF model

We've already discussed free cash flow models, so now let's get to the hard work: actually calculating value using these models.

Example: Calculating firm value with a single-stage FCFF model

Knappa Valley Winery's (KVW) most recent FCFF is \$5,000,000. KVW's target debt-to-equity ratio is 0.25. The market value of the firm's debt is \$10,000,000, and KVW has 2,000,000 shares of common stock outstanding. The firm's tax rate is 40%, the shareholders require a return of 16% on their investment, the firm's before-tax cost of debt is 8%, and the expected long-term growth rate in FCFF is 5%. Calculate the value of the firm and the value per share of the equity.

Answer:

According to the firm valuation formula using the Single-stage FCFF model, we have:

$$V_f = \frac{FCFF_0(1 + g)}{WACC - g}$$

We are given $FCFF_0 = \$5,000,000$, $g = 5\%$. Hence, we need to calculate WACC as follows:

$$WACC = w_e \times r_e + w_d \times r_d \times (1 - \text{tax rate})$$

$$= 0.8 \times 0.16 + 0.20 \times 0.08 \times (1 - 0.40) = 0.1376 = 13.76\%.$$

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

1.

Single-stage FCFF model

Example: Calculating firm value with a single-stage FCFF model

Answer:

Now, we can calculate the value of the firm as:

$$V_f = \frac{FCFF_0(1+g)}{WACC - g} = \frac{\$5,000,000 \times (1 + 0.05)}{0.1376 - 0.05} = \$59,931,507$$

Given that debt is worth \$10,000,000, the implied total value of the equity is:

$$V_e = V_f - V_d = \$59,931,507 - \$10,000,000 = \$49,931,507$$

Now, with 2,000,000 shares outstanding, the value of the equity per share is:

$$V_{e/s} = \frac{V_e}{\text{Shares outstanding}} = \frac{\$49,931,507}{2,000,000} = \$24.97.$$

Note: The actual debt-to-equity ratio ($10,000,000 / 49,931,507 = 0.20$) does not equal the target ratio of 0.25. There is nothing inconsistent in this example, because WACC is usually calculated using target capital weights.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

2.

Single-stage FCFE model

Example: Calculating equity value with a single-stage FCFE model

Ridgeway Construction has an FCFE of 2.50 Canadian dollars (C\$) per share and is currently operating at a target debt-to-equity ratio of 0.4. The expected return on the market is 9%, the risk-free rate is 4%, and Ridgeway has a beta of 1.5. The expected growth rate of FCFE is 4.5%. Calculate the value of Ridgeway stock.

Answer:

According to the equity valuation formula using the Single-stage FCFE model, we have:

$$V_e = \frac{FCFE_0(1 + g)}{r - g}$$

We are given $FCFE_0 = \text{C\$}2.5$, $g = 4.5\%$. Hence, we need to calculate r with CAPM as follows:

$$\begin{aligned} r &= \text{risk-free rate} + \text{beta} \times (\text{expected return on market} - \text{risk-free rate}) \\ &= 0.04 + 1.50 \times (0.09 - 0.04) = 0.115 = 11.5\% \end{aligned}$$

Now, we can calculate the value of the equity per share as:

$$V_e = \frac{FCFE_0(1 + g)}{r - g} = \frac{\text{C\$}2.5 \times (1 + 0.045)}{0.115 - 0.045} = \text{C\$}37.32.$$

[LOS 3.I] Describe approaches for calculating the terminal value in a multistage valuation model

1.

Calculating the terminal value

There are **two** basic approaches for calculating terminal value:

Using the single-stage model

We forecaste an FCFF or FCFE at the point in time at which cash flows begin to grow at the long-term, stable growth rate, and then we estimated terminal value using a single-stage model.

Terminal value using SSM

$$V_{ft} = \frac{FCFF_t(1+g)}{WACC - g}$$

$$V_{et} = \frac{FCFE_t(1+g)}{r - g}$$

Using the market multiple approach

We use market price multiples, especially a P/E ratio at the forecast horizon and then estimate the terminal value as the P/E multiplied by the earnings estimate.

Terminal value using MM

$$V_t = \frac{P_t}{E_t} \times E_t$$

$$\text{or } V_t = \frac{P_t}{E_{t+1}} \times E_{t+1}$$

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

3.

Two-stage FCFF model

Example: Calculating equity value with a single-stage FCFE model

The Prentice Paint Company earned a net profit margin of 20% on revenues of \$20 million this year. Fixed capital investment was \$2 million, and depreciation was \$3 million. Working capital investment equals 7.5% of sales every year. Net income, fixed capital investment, depreciation, interest expense, and sales are expected to grow at 10% per year for the next five years. After five years, the growth in sales, net income, fixed capital investment, depreciation, and interest expense will decline to a stable 5% per year. The tax rate is 40%, and Prentice has 1 million shares of common stock outstanding and long-term debt paying 12.5% interest trading at its par value of \$32 million. Calculate the value of the firm and its equity using the FCFF model if the WACC is 17% during the high-growth stage and 15% during the stable stage.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

3.

Two-stage FCFF model

Answer:

Let's demonstrate the calculation of the FCFF in Year 0 (in per-share amounts):

$$\text{Net income} = \$20.00 \times 0.20 = \$4.00$$

$$\text{Interest} = \$32.00 \times 0.125 = \$4.00$$

$$\text{Interest} \times (1 - \text{tax rate}) = \$4.00 \times (1 - 0.40) = \$2.40$$

$$\text{WCInv} = \$20.00 \times 0.075 = \$1.50$$

We are given Sales = \$20.00, Depreciation = \$3, and FCInv = \$2.00.

$$\text{Thus FCFF} = \text{NI} + \text{Dep} + \text{Int} \times (1 - t) - \text{FCInv} - \text{WCInv}$$

$$= \$4.00 + \$3.00 + \$2.40 - \$2.00 - \$1.50 = \$5.90.$$

Based on the growth rate at 10% per year for the next five years, similarly, we have the components of FCFF are calculated in the following table (see next slide).

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

3.

Two-stage FCFF model

Answer:

The components of FCFF are calculated in the following table:

FCFF for Years 0 Through 6 (in per-share amounts of \$):

	0	1	2	3	4	5	6
Sales (\$)	20.00	22.00	24.20	26.62	29.28	32.21	33.82
NI	4.00	4.40	4.84	5.32	5.86	6.44	6.76
Int x (1-t)	2.40	2.64	2.90	3.19	3.51	3.87	4.06
Dep	3.00	3.30	3.63	3.99	4.39	4.83	5.07
FCInv	2.00	2.20	2.42	2.66	2.93	3.22	3.38
WCInv	1.50	1.65	1.82	2.00	2.20	2.42	2.54
FCFF	\$5.90	\$6.49	\$7.13	\$7.84	\$8.63	\$9.50	\$9.97

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

3.

Two-stage FCFF model

Answer:

In Year 1, sales grow by 10% to \$22.00 per share. Following five years of 10% growth, the growth of each component falls to 5%.

The terminal value (as of Year 5, discounted at the stable WACC of 15%) is:

$$\text{Terminal value} = \frac{\text{FCFF}_6}{\text{WACC} - g} = \frac{\$9.97}{0.15 - 0.05} = \$99.70$$

We can place the cash flows to be evaluated on a time line, such as the one in the following figure, to get a clearer picture of what we need to evaluate.

FCFF Timeline:



Notice that the WACC in the high-growth stage (17%) is different than the stable stage (15%).

We calculated terminal value in Year 5 using 15%, but we'll calculate the present value today of the high-growth cash flows and the terminal value at 17%. The total of the firm today is:

$$V_f = \frac{\$6.49}{1.17} + \frac{\$7.13}{1.17^2} + \frac{\$7.84}{1.17^3} + \frac{\$8.63}{1.17^4} + \frac{\$109.20}{1.17^5} = \$70.06.$$

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

3.

Two-stage FCFF model

Answer:

FCFF Timeline:



To perform this calculation quickly and accurately, use the following keystrokes on your financial calculator:



CF0 = 0

C01 = 6.49

C02 = 7.13

C03 = 7.84

C04 = 8.63

C05 = 109.20

I = 17; CPT → NPV = \$70.06.

Thus, given that the value of the firm's debt is \$32 per share, the value of equity per share is \$70.06 – \$32.00 = \$38.06.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

4.

Two-stage FCFE model

Example: Two-stage FCFE model with declining growth in Stage 1

Suppose that Sioux Falls Decor has revenues of \$20 million this year. We assume that its future performance will be tracked relative to sales as follows:

Sales and Net Margin Forecasts

Year	1	2	3	4	5	6
Sales growth	30%	25%	20%	15%	10%	5%
Net profit margin	8.0%	7.5%	7.0%	6.0%	5.5%	5.0%

- Fixed capital investment net of depreciation is projected to be 30% of the sales increase in each year.
- Working capital requirements are 7.0% of the projected dollar increase sales in each year.
- Debt will finance 40% of the investments in net capital and working capital.
- The company has a 12% required rate of return on equity.
- The firm has 1 million shares of common stock outstanding.

Calculate the value of the equity of Sioux Falls using the two-stage FCFE model .

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

4.

Two-stage FCFE model

Answer:

We recognize that the target debt-to-asset ratio (DR) is 0.40.

Let's demonstrate the calculation of the cash flow components in Year 1 (in per-share amounts):

$$\text{Sales} = \$20.00 \times 1.30 = \$26.00$$

$$\text{Net income} = \$26.00 \times 0.08 = \$2.08$$

$$\text{Net FCInv} = (\$26.00 - \$20.00) \times 0.3 = \$1.80$$

$$\text{WCInv} = (\$26.00 - \$20.00) \times 0.07 = \$0.42$$

$$\begin{aligned} \text{FCFE} &= \text{NI} - (1 - \text{DR}) \times (\text{FCInv} - \text{Dep}) - (1 - \text{DR}) \times \text{WCInv} \\ &= \$2.08 - (1 - 0.4) \times \$1.80 - (1 - 0.4) \times \$0.42 = \$0.748 \end{aligned}$$

$$\text{Note: Debt financing} = \text{DR} \times (\text{FCInv} - \text{Dep} + \text{WCInv})$$

$$= 0.4 \times (\$1.80 + \$0.42) = \$0.888$$

Similarly, we have the following table that shows the FCFE for Years 1 through 6 (\$ amounts are per share) (see next slide)

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

4.

Two-stage FCFE model

Answer:

Calculating FCFE for Years 1 Through 6:

Year	1	2	3	4	5	6
Sales growth	30%	25%	20%	15%	10%	5%
Net profit margin	8.0%	7.5%	7.0%	6.0%	5.5%	5.0%
Sales	\$26.00	\$32.50	\$39.00	\$44.85	\$49.335	\$51.802
NI	2.08	2.44	2.73	2.691	2.71	2.59
FCInv - Dep	1.80	1.95	1.95	1.755	1.346	0.74
WCInv	0.42	0.455	0.455	0.4095	0.314	0.171
Debt financing	0.888	0.962	0.962	0.864	0.664	0.364
FCFE	\$0.748	\$0.997	\$1.287	\$1.391	\$1.714	\$2.043

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

4.

Two-stage FCFE model

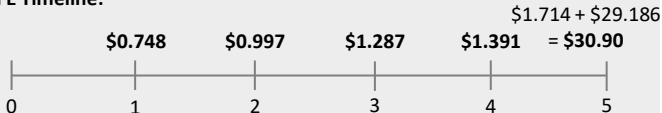
Answer:

Terminal value (as of Year 5, assuming 5% stable long-term growth) is equal to:

$$\text{Terminal value} = \frac{\text{FCFE}_6}{r - g} = \frac{\$2.043}{0.12 - 0.05} = \$29.186$$

We can place the cash flows to be evaluated on a time line, such as the one in the following figure, to get a clearer picture of what we need to evaluate.

FCFE Timeline:



Total current value of equity:

$$V_e = \frac{\$0.748}{1.12} + \frac{\$0.997}{1.12^2} + \frac{\$1.287}{1.12^3} + \frac{\$1.391}{1.12^4} + \frac{\$30.90}{1.12^5} = \$20.80$$

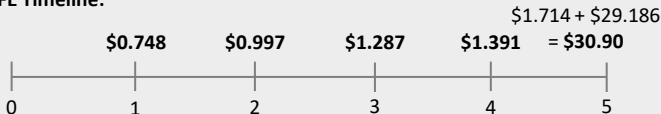
[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

4.

Two-stage FCFE model

Answer:

FCFE Timeline:



As usual, we would rely on the cash flow keys of our financial calculator to perform the previous calculation:



CF0 = 0

C01 = 0.748

C02 = 0.997

C03 = 1.287

C04 = 1.391

C05 = 30.90

I = 12; CPT → NPV = \$20.80.

Thus, the value of equity per share is \$20.80.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

5.

Three-stage FCFE model

Example: Three-stage FCFE model with forecast growth in total FCFE

Medina Classic Furniture, Inc., is expected to experience growth in three distinct stages in the future. Its most recent FCFE is 0.90 Canadian dollars (C\$) per share.

The following information has been compiled:

High-growth period:

- Duration = 3 years.
- FCFE growth rate = 30%.
- Shareholders' required return = 20%.

Transitional period:

- Duration = 3 years.
- FCFE growth will decline by 9% per year down to the indicated stable growth rate.
- Shareholders' required return = 15%.

Stable-growth period:

- FCFE growth rate = 3%
- Shareholders' required return = 10%.

Calculate the value of the firm's equity using the three-stage FCFE model.

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

5.

Three-stage FCFE model

Example: Three-stage FCFE model with forecast growth in total FCFE

Answer:

The annual FCFE and the associated present value are presented in the table:

FCFE and PV

High-growth period	Year 1	Year 2	Year 3
Growth rate	30%	30%	30%
FCFE	C\$1.170	C\$1.521	C\$1.977
PV ($r = 20\%$)	C\$0.975	C\$1.056	C\$1.144
Transition period	Year 4	Year 5	Year 6
Growth rate	21%	12%	3%
FCFE	C\$2.393	C\$2.680	C\$2.760
PV	C\$1.204	C\$1.173	C\$1.050

[LOS 3.k] Estimate a company's value using the appropriate free cash flow model(s)

5.

Three-stage FCFE model

Example: Three-stage FCFE model with forecast growth in total FCFE

Answer:

The transitional present values are computed using a combination of the 20% initial discount rate and the transitional 15% rate. For example, the present value of $FCFE_5$ is computed as: $\frac{2.680}{1.20^3 \times 1.15^2} = C\1.173 .

We can calculate the terminal value of the stock as of Year 6 using the FCFE projected for Year 7. Notice that we use the Stage 3 required return of 10%.

$$\text{Terminal value} = \frac{2.760 \times 1.03}{0.10 - 0.03} = C\$40.611$$

Thus, the value of Medina stock is:

$$V_e = 0.975 + 1.056 + 1.144 + 1.204 + 1.173 + 1.050 + \frac{40.611}{1.20^3 \times 1.15^3} = C\$22.055.$$

Note: Due to the changing discount rates, our financial calculator was not as helpful as it was in other multiple cash flow calculations.

[LOS 3.m] Evaluate whether a stock is overvalued, fairly valued, or undervalued based on a free cash flow valuation model

1.

Evaluating value of a stock

We compare the price at which the stock is trading (Market price) and the price implied by a free cash flow valuation model (Model price):

- Market price > Model price → Overvalued
- Market price < Model price → Undervalued
- Market price = Model price → Fairly valued